

NBER WORKING PAPER SERIES

WHEN PUBLIC SIGNALS BACKFIRE:
STRATEGIC DISCLOSURE AND INFORMATION CROWD-OUT

Marty Haoyuan Chen
Ginger Zhe Jin

Working Paper 35625
<http://www.nber.org/papers/w35625>

NATIONAL BUREAU OF ECONOMIC RESEARCH
1050 Massachusetts Avenue
Cambridge, MA 02138
August 2026

We would like to thank Tingting Ding, Agata Farina, Emel Filiz-Ozbay, Sueyoul Kim, Mario Leccese, Sanxi Li, Bin Miao, Erkut Ozbay, Harry Pei, Tri Phu Vu, and seminar participants at Tsinghua University, Renmin University of China, and University of Maryland for helpful comments and discussions. This project was approved by the IRB at the University of Maryland. The views expressed herein are those of the authors and do not necessarily reflect the views of the National Bureau of Economic Research.

NBER working papers are circulated for discussion and comment purposes. They have not been peer-reviewed or been subject to the review by the NBER Board of Directors that accompanies official NBER publications.

© 2026 by Marty Haoyuan Chen and Ginger Zhe Jin. All rights reserved. Short sections of text, not to exceed two paragraphs, may be quoted without explicit permission provided that full credit, including © notice, is given to the source.

When Public Signals Backfire: Strategic Disclosure and Information Crowd-Out
Marty Haoyuan Chen and Ginger Zhe Jin
NBER Working Paper No. 35625
August 2026
JEL No. C91, D82, D83, D91, L15

ABSTRACT

We study voluntary disclosure when partially naive receivers observe a public signal correlated with the sender's private information. We develop a framework in which the public signal changes how receivers interpret silence, and derive a closed-form disclosure threshold nesting classical unraveling. A laboratory experiment varying the correlation confirms our predictions: disclosure decreases as the signal becomes more favorable, and receiver guesses upon nondisclosure rise toward it. The public signal can crowd out disclosure, leaving receivers less informed than without it. This informational loss peaks at intermediate correlation, where the signal shifts beliefs yet remains too noisy to substitute for disclosure.

Marty Haoyuan Chen
University of Maryland, College Park
hchen915@umd.edu

Ginger Zhe Jin
Boston University
Questrom School of Business
Markets, Public Policy & Law
and NBER
gzjin@bu.edu

1 Introduction

In many settings, buyers form expectations about product quality from two informational sources: public signals, and private information held by the seller, who can disclose or withhold it but cannot fabricate it. A bond issuer, for example, may privately solicit a credit rating while investors already observe competing ratings, market prices, and audited financial statements;¹ a restaurateur may broadcast or withhold the latest sanitation inspection result while customer reviews and prior grades remain public;² an HMO or a hospital observes its current quality and accreditation scores besides prior public disclosures and peer reporting decisions;³ a firm collects internal environmental data alongside public emissions databases and peer disclosures;⁴ and a college applicant may or may not report a standardized test score to test-optional colleges, while grade-point average, course rigor, recommendation letters, and high-school quality are required in the application.⁵

In each case the public and private signals are correlated, and the buyer must form beliefs about the private information from the public signals she already sees together with her reading of the seller’s disclosure choice. Correlated public signals and discretionary verifiable disclosure thus coexist as the rule in many real-world markets, not the exception. This poses an immediate question for policymakers: can we trust the market to disclose the private state voluntarily, assuming consumers fully understand what the disclosed information conveys alongside the public signal?

At first glance, the classical unraveling literature answers this cleanly. [Grossman and Hart \(1980\)](#), [Grossman \(1981\)](#), and [Milgrom \(1981\)](#) show that, in the unique sequential equilibrium of a verifiable disclosure game with a sufficiently rich state space, all but the worst types disclose and silence is read as the worst news—regardless of whether a public signal exists or how it correlates with the private state. A large empirical and experimental literature, however, finds that this benchmark fails dramatically: senders withhold a substantial fraction of intermediate types, and receivers are too generous toward silence ([Dranove and Jin, 2010](#); [Jin, Luca and Martin, 2021](#)). Most of that literature treats the receiver’s information environment as fixed, asking how senders

¹To address bond issuers’ secret rating shopping behavior, India’s Securities and Exchange Board (SEBI) tightened the rules in 2016 to require credit rating agencies to disclose ratings even when issuers did not accept them; agencies such as ICRA now maintain public lists of “ratings assigned and not accepted”, and the U.S. Securities and Exchange Commission (SEC) has separately considered mandatory disclosure of preliminary ratings ([Mählmann, 2008](#); [Kallapur et al., 2025](#)).

²[Gurney and Loewenstein \(2020\)](#) show that restaurant patrons fail to discount missing sanitation inspection grades by the right amount, even when interventions are designed to make the missing information more salient.

³[McCormick et al. \(2002\)](#) document that of the 329 HMOs that publicly disclosed HEDIS scores in 1997, 161 (49 percent) withdrew from public disclosure in 1998, with plans in the lowest-quality tertile far more likely to withdraw than plans in the highest-quality tertile. [Ito and Sugawara \(2005\)](#) find that hospital accreditation scores are positively associated with the public disclosure of accreditation reports.

⁴[Marquis, Toffel and Zhou \(2016\)](#) show that selective greenwashing is moderated by civil-society scrutiny and transparency norms. [Ahn, Cho and Cho \(2025\)](#) find that the negative stock-market reaction to disappointing environmental performance is smaller for high-reputation firms.

⁵Recent evidence shows that applicants withhold scores selectively for test-optional schools and condition disclosure on their other observable attributes and on the school’s stated leniency toward non-reporting ([McManus, Howell and Hurwitz, 2023](#); [Chen and Jin, 2025](#)).

behave when the message itself is all receivers see. It thus largely sets aside the public signals buyers actually observe in the field, and skips the question of how voluntary disclosure responds when those signals are correlated with the private state.

To fill this gap, we take up two questions. How does voluntary disclosure respond to a correlated public signal when receivers are partially naive? And when does introducing that signal—market-generated or policy-mandated—improve or worsen the overall information environment? We answer both by combining a novel theoretical framework with a laboratory experiment.

On the theory side, we extend the classical sender-receiver disclosure game by letting nature draw a public signal θ_b , observed by both parties and correlated with the sender’s private state θ_a with correlation parameter ρ . Crucially, we allow the receiver’s belief upon non-disclosure to be a convex combination of the rational non-disclosure mean and the public signal itself, with weight λ_ρ on the public signal increasing in the correlation. This captures the idea that, when a salient and easily accessible reference point is available, partially naive receivers anchor on it rather than fully working out the strategic content of silence. The framework nests two limiting cases: at $\lambda_\rho = 0$, full unraveling obtains and the equilibrium reduces to the classical Milgrom benchmark; with no public signal, the model collapses to the disclosure game with receiver naivety studied in [Jin, Luca and Martin \(2021\)](#). Between these limits, the unique equilibrium features a disclosure threshold strictly increasing in the public signal whenever the signal exceeds the worst possible state, and strictly increasing in ρ .

The departure from full unraveling decomposes additively into a baseline component independent of the public signal (the failure of classical unraveling) and a signal-contingent component proportional to the gap between the public signal and the prior mean. The signal-contingent term is the source of information crowd-out when the public signal is favorable and crowd-in when it is unfavorable. Solving for informativeness in a numerical example—where informativeness is the receiver’s expected payoff—we further map regions of the (θ_b, ρ) space in which adding a public signal strictly enhances or reduces the informativeness of the disclosure environment.

Intuitively, the strength of the public signal, measured by its correlation with the private state (ρ), affects disclosure incentives *non-linearly*. At the extremes, disclosure incentives are either independent of the public signal by definition ($\rho = 0$) or irrelevant because consumers read the private state directly off a perfectly correlated signal ($\rho = 1$). In the middle range, where the public signal carries meaningful but noisy information, receivers’ tendency to anchor on it invites the sender to hide behind a favorable signal, blurring the informativeness of everything receivers see (that is, public signal plus strategic silence). As ρ rises, the private state becomes easier to infer from the public signal, but the countervailing incentive to conceal a mediocre state behind a favorable one rises as well. When crowd-out is strong enough, overall transparency is lower with the public signal than without it.

On the experimental side, we ran 50-round disclosure games with 238 subjects across low-, medium-, and high-correlation treatments ($\rho \in \{0.2, 0.5, 0.8\}$), holding the state space, payoff functions, and information protocol fixed. Three findings stand out. First, voluntary disclosure

declines as the public signal becomes more favorable, with the decline concentrated at intermediate private states and largest under the highest correlation. Second, receiver guesses upon non-disclosure rise systematically with the public signal and with the correlation, exceeding the rational benchmark as the model predicts. Estimating the naivety parameter by treatment yields $\hat{\lambda}_{0.2} = 0.17$, $\hat{\lambda}_{0.5} = 0.26$, and $\hat{\lambda}_{0.8} = 0.32$ —monotone in ρ and well below the corresponding correlation parameter, so receivers are partially but not fully naive. Applied to the no-public-signal benchmark of [Jin, Luca and Martin \(2021\)](#), the same estimator yields $\hat{\lambda}_0 \approx 0.16$, just below our low-correlation estimate and consistent with the monotone pattern.⁶ The estimated model fits substantially better than full unraveling ($\lambda = 0$), full naivety ($\lambda = 1$), and the cursed equilibrium of [Eyster and Rabin \(2005\)](#). Third, when the public signal is favorable, inference accuracy upon non-disclosure falls below the no-public-signal benchmark at all three values of ρ . Once we account for the information gain in disclosed cases, overall informativeness remains below the benchmark under low and medium correlation, with the largest loss at the intermediate ρ , and recovers under high correlation, where near-universal disclosure offsets the poor inference from silence. The informational loss is therefore non-monotone in the correlation, as the theory predicts.

In short, the public signal can backfire, especially where naivety is greatest and the sender’s incentive to hide behind the signal is strongest. When receivers are not fully sophisticated, a public signal becomes a salient cognitive reference point for interpreting silence. A favorable signal pulls a naive receiver’s reading of silence upward, weakening the sender’s incentive to disclose intermediate states, expanding the non-disclosure pool, and reducing the total information receivers extract from silence. When strategic withholding and its naive interpretation are strong enough, disclosure behavior can crowd out the public signal entirely in the informativeness of non-disclosure, particularly when the correlation between public signal and private state is substantial but not exceedingly high.

These results carry policy and business implications well beyond the lab; we highlight three. First, because disclosure incentives depend on the public signals already observable to the market, policies on whether to mandate or standardize disclosure of a private state must be conditioned on existing public signals and their correlation with that state. Second, because crowding out and crowding in arise only when receivers misread silence, consumers, investors, patients, and college admissions officers should be educated about the strategic incentives behind non-disclosure and the role public signals play in shaping them. Third, although our study does not address why the public signal exists—it may come from mandatory disclosure laws (e.g., SEC financial disclosure requirements), firms’ voluntary disclosure in earlier periods (e.g., a CEO’s earnings forecast alongside mandatory filings), or third-party reports (e.g., academic research or industry association reports)—whoever releases a public signal must weigh its direct value against firms’ stronger incentive to withhold private states down the road when the public ones are favorable. These dynamic considerations can produce a complex strategic game among firms, regulators, and related

⁶The qualitative no-public-signal benchmark (i.e., substantial incomplete unraveling and receivers’ insufficient skepticism toward nondisclosure), is well established across closely related laboratory disclosure studies. We use [Jin, Luca and Martin \(2021\)](#) for the quantitative comparison because its experimental protocol is closest to ours.

third parties.

The remainder of the paper is organized as follows. Section 2 reviews the literature and articulates our contribution. Section 3 presents the theoretical framework and main equilibrium results. Section 4 describes the experimental design, Section 5 reports the empirical findings, and Section 6 concludes with real-world implications and directions for future research.

2 Related Literature

Our paper contributes to several strands of the disclosure literature. The first is the empirical literature on voluntary disclosure that documents the failure of full unraveling and the role of receiver behavior (Dranove and Jin, 2010; Feltoovich, Harbaugh and To, 2002; Hirshleifer and Teoh, 2003; Eyster and Rabin, 2005; Board, 2009; McManus, Howell and Hurwitz, 2023). Most of this literature treats the receiver’s information environment as fixed and asks how senders behave; we instead vary the informational environment by introducing a correlated public signal and ask how voluntary disclosure responds. Sector-specific evidence on selective disclosure—solicited credit ratings (Mählmann, 2008; Kallapur et al., 2025), management earnings forecasts (Skinner, 1994), IPO prospectus forecasts (Buchner, Mohamed and Saadouni, 2017), environmental disclosures (Marquis, Toffel and Zhou, 2016; Ahn, Cho and Cho, 2025), restaurant hygiene grading (Gurney and Loewenstein, 2020), and hospital and HMO quality reporting (McCormick et al., 2002; Ito and Sugawara, 2005)—documents the phenomenon in many markets but rarely treats the public signal as the primary object of analysis. We put the public signal at the center, derive how it shapes voluntary disclosure incentives, and identify the conditions under which it enhances or backfires.

The second strand is the experimental literature on disclosure (Jin, Luca and Martin, 2021, 2022; Penczynski, Koch and Zhang, 2022; Farina and Leccese, 2024; Farina et al., 2024; Hagenbach and Perez-Richet, 2018; Montero and Sheth, 2021; Chen and Jin, 2025). Jin, Luca and Martin (2021) is closest in protocol; we adopt their state space and payoff functions, and our low-correlation naivety estimate lies just above the no-public-signal estimate implied by their data. Our contribution relative to this literature is to systematically vary the public signal and its correlation with the private state, document information crowd-out as a within- and across-treatment phenomenon, estimate how naivety scales with correlation, and characterize when the public signal causes overall informativeness to backfire.

A third strand is the theoretical literature on voluntary disclosure with disclosure costs (Jovanovic, 1982; Verrecchia, 1983) and informational frictions (Dye, 1985; Jung and Kwon, 1988; Beyer et al., 2010; Cheynel, 2013; Kim, Taylor and Verrecchia, 2021; Aghamolla and An, 2021). Our closest antecedents are Acharya, DeMarzo and Kremer (2011), Quigley and Walther (2020), and Frenkel, Guttman and Kremer (2020). Acharya, DeMarzo and Kremer (2011) study a dynamic disclosure game with a Dye-type friction (see also Guttman, Kremer and Skrzypacz, 2014) in which investors are rational but do not know whether or when the firm is informed, so silence is not fully revealing and disclosure timing responds to external news: adverse public news

triggers immediate disclosure, while favorable news delays it. Their prediction that favorable public news slows disclosure parallels our crowd-out result, but the mechanisms differ sharply, as they rely on rational dynamics and preemptive disclosure. [Quigley and Walther \(2020\)](#) study a disclosure game with a correlated public signal in banking, in a setting that relies on disclosure costs and preemptive disclosure, with the sender committing to a strategy before observing the signal. [Frenkel, Guttman and Kremer \(2020\)](#) likewise study voluntary disclosure alongside a correlated third-party information source, and show that although outside information crowds out disclosure by the firm, aggregate information improves. In contrast to all three, our framework generates non-unraveling through receiver naivety, without disclosure costs, information dynamics, or ex-ante disclosure commitment, and lets the sender condition disclosure on the realized signal; this yields signal-contingent withholding as a comparative-static prediction and a closed-form decomposition of the disclosure threshold into a baseline and a signal-contingent component that we take to the data.

The contrast with fully rational receivers also clarifies what naivety adds. [Rappoport \(2025\)](#) characterizes how changes in the receiver’s prior over the sender’s evidence induce more or less rational skepticism toward silence; we study the analogous comparative static when the prior shifts with a realized public signal and receivers are only partially skeptical. In [Frenkel, Guttman and Kremer \(2020\)](#), the crowd-out never overturns the direct informational gain from the outside source, whereas receiver naivety in our setting allows it to. The finding that public information can reduce welfare when agents overweight it echoes [Morris and Shin \(2002\)](#), though their mechanism operates through coordination motives among fully rational agents rather than naive inference from silence.

A fourth strand connects to behavioral models of receivers in disclosure environments. [Eyster and Rabin \(2005\)](#) introduce cursed equilibrium, in which receivers underappreciate the dependence of senders’ messages on their private information, a primitive that [Fong, Lin and Palfrey \(2025\)](#) and [Cohen and Li \(2026\)](#) extend to multi-stage games of incomplete information, characterizing how such neglect distorts belief updating along the history of play and how players draw inference from observed relative to hypothetical events. Our specification departs from the cursed formulation in how naivety is parameterized: rather than partial neglect of the conditional expectation $E[\theta_a \mid \theta_b]$, we model naivety as anchoring on the observable signal θ_b itself, a more cognitively parsimonious heuristic. In our data, the simpler heuristic fits more closely than the cursed alternative in seven of the nine treatment-signal cells, with an average absolute deviation roughly half as large, suggesting that receivers’ departures from Bayesian inference of silence are best captured by anchoring on the simplest available signal rather than by partial neglect of strategic content.

Our formulation also connects naivety about silence to well-documented failures of statistical reasoning. [Enke \(2020\)](#) shows that subjects estimating a state from selected evidence often follow a heuristic of averaging the visible data while the unobserved signals never come to mind, and that neglect is not an exogenously given parameter but varies with the complexity of the decision

problem and with whether attention is drawn to the presence of selection. [Farina et al. \(2024\)](#) and [Farina and Leccese \(2024\)](#) document the same failure on the receiver side of disclosure games, where subjects respond too optimistically to evidence that a sender has concealed or selected, consistent with broader evidence on the difficulty of contingent reasoning ([Martínez-Marquina, Niederle and Vespa, 2019](#); [Esponda and Vespa, 2024](#)). [Enke and Zimmermann \(2019\)](#) describe a further failure of this kind: when signals share a common source, subjects treat them as independent and double count the information the source contributes. Beliefs are then excessively sensitive to the common source and overshoot in its direction, even though they remain unbiased in expectation.

Related work shows how receiver naivety can support non-disclosure as an equilibrium outcome, including through shrouded attributes ([Gabaix and Laibson, 2006](#); [Heidhues, Kőszegi and Murooka, 2016](#)) and coarse thinking ([Mullainathan, Schwartzstein and Shleifer, 2008](#)), and [Hirshleifer and Teoh \(2003\)](#) document analogous frictions in financial reporting. A broader literature documents conservatism in belief updating ([Edwards, 1968](#); [Benjamin, 2019](#)), whereby receivers underweight new information relative to a salient prior or reference point.

3 Theoretical Framework

3.1 The Disclosure Game and Classical Unraveling

A sender (S) and a receiver (R) interact in a one-shot disclosure game. Before the game begins, nature draws a public signal $\theta_b \in \mathcal{B}$ according to a known distribution, and both players observe its realization. Nature then draws a private state $\theta_a \in \mathcal{A}$ according to a known conditional distribution $\pi(\theta_a|\theta_b)$, such that θ_a and θ_b are positively correlated. Only the sender observes the realization of the private state. Thus, the sender observes both θ_a and θ_b , while the receiver observes only θ_b and the sender's subsequent disclosure decision.

The sender has two possible actions: either report θ_a to the receiver or send no message. This report must be truthful (i.e. the sender cannot misrepresent θ_a). Thus, the sender's action set is $M(\theta_a) = \{\theta_a, \emptyset\}$. The receiver observes (θ_b, m) where $m \in \{\theta_a, \emptyset\}$, and chooses an action $\hat{\theta}_a$ from a set $\mathcal{G}$, which contains $\mathcal{A}$. The receiver action is interpreted as a guess of the true state θ_a .

The true state θ_a and the receiver's guess jointly determine payoffs for both parties. The sender's payoff is given by a function $U_S(\hat{\theta}_a)$, which is strictly increasing in the receiver's guess and independent of θ_a . The receiver's payoff is given by a function $U_R(\hat{\theta}_a, \theta_a)$, which is concave in $\hat{\theta}_a$ and reaches its maximum when $\hat{\theta}_a = \theta_a$. In other words, the sender always benefits from higher receiver guesses regardless of the true private state, while the receiver benefits from selecting an action closer to the true private state. These payoff functions produce a strong conflict of interest, especially when θ_a is low relative to θ_b : in such cases the sender may be incentivized to withhold, depending on how the receiver's perception of a non-disclosed θ_a varies by the public signal θ_b .

When the state space $\mathcal{A}$ is sufficiently rich, the unique sequential equilibrium exhibits full unraveling: the sender discloses all states except the minimum element of $\mathcal{A}$, and the receiver

interprets non-disclosure as the lowest state. This follows from the classic unraveling result of [Milgrom \(1981\)](#). Importantly, this prediction is invariant to the realization of θ_b and to the correlation between θ_a and θ_b . Although the public signal affects the receiver’s prior, verifiable disclosure eliminates the possibility of misrepresentation, so the unraveling logic applies regardless of the information structure.

3.2 Equilibrium Predictions with Receiver Naivety

Given the widespread failure of the classical unraveling result in the lab and in real-world examples ([Jin, Luca and Martin, 2021](#); [Dranove and Jin, 2010](#)), this subsection allows receiver naivety in the disclosure game and derives equilibrium predictions under such receiver naivety.

For ease of illustration, we impose several assumptions on the state space and the information structure. First, we assume the set $\mathcal{A}$ is a non-degenerate interval, $\mathcal{A} = [\theta_a^{min}, \theta_a^{max}]$ with $\theta_a^{max} > \theta_a^{min}$. Second, let $\mathcal{B} = \mathcal{A}$ such that the public signal is drawn from the same range of the private signal. Third, conditional on the public signal θ_b , with probability $\rho \in (0, 1)$ the private state equals the public signal θ_b , and with probability $1 - \rho$ it is drawn uniformly from $\mathcal{A}$. The parameter ρ thus governs the strength of the correlation between the private state and the public signal: higher values of ρ correspond to a public signal that is more predictive of θ_a .

In equilibrium, the sender’s optimal disclosure strategy is characterized by a cutoff rule:

$$\sigma(\theta_a, \theta_b) = \mathbb{1}[\theta_a > \theta_a^*(\theta_b)] \quad (1)$$

where $\theta_a^*(\theta_b)$ is the disclosure threshold.

Importantly, we allow the receiver’s belief formation following non-disclosure to depart from full rationality. In particular, we model the receiver’s guess upon non-disclosure as a weighted average of the mean of the non-disclosure pool and the public signal:

$$\hat{\theta}_a(\emptyset, \theta_b) = (1 - \lambda_\rho) \cdot \mathbb{E}[\theta_a | \theta_a \leq \theta_a^*(\theta_b), \theta_b] + \lambda_\rho \cdot \theta_b \quad (2)$$

where $\lambda_\rho \in [0, 1)$ is the degree of naivety, which depends on the correlation parameter ρ . We do not make any assumption on the functional form of the weight (λ_ρ) that the receiver puts on the public signal, but assume that λ_ρ increases in ρ : a more predictive public signal serves as a more salient cognitive reference, and thus induces greater naivety.

This specification captures the idea that the public signal θ_b serves as a salient and cognitively accessible reference point, and that the receiver insufficiently adjusts away from this reference point after observing information withholding. Thus, using θ_b itself as the reference point reflects a form of naivety in the updating of beliefs, and the weight on this reference point (λ_ρ) captures the degree of naivety. As a result, non-disclosure is perceived as less indicative of low realizations of θ_a than in the fully rational benchmark, and the degree of naivety (λ_ρ) may itself depend on how the public signal correlates with the sender’s private information (ρ).

Proposition 1 (Uniqueness of equilibrium). *For every θ_b , there exists a unique equilibrium disclosure threshold, given by*

$$\theta_a^*(\theta_b) = \frac{1 - \lambda_\rho}{1 + \lambda_\rho} \theta_a^{min} + \frac{2\lambda_\rho}{1 + \lambda_\rho} \theta_b. \quad (3)$$

In particular, $\theta_a^(\theta_a^{min}) = \theta_a^{min}$ and $\theta_a^*(\theta_b) < \theta_b$ for all $\theta_b > \theta_a^{min}$.⁷*

Corollary 1. *Suppose $\lambda_\rho = 0$, then the unique equilibrium threshold satisfies $\theta_a^*(\theta_b) = \theta_a^{min}$ for all θ_b .*

When $\lambda_\rho = 0$, the sender discloses all states except the minimum, and the receiver correctly infers the minimum upon non-disclosure, recovering full unraveling. Thus, our model embeds the sequential equilibrium with full rationality as a special case.

Corollary 2. *Suppose $\lambda_\rho \in (0, 1)$, then $\theta_a^*(\theta_b) > \theta_a^{min}$ for all $\theta_b > \theta_a^{min}$, and $\theta_a^*(\theta_b)$ is strictly increasing in θ_b .*

When the receiver puts some positive weight on the public signal, it introduces optimism beyond the most pessimistic belief that a rational receiver should hold for non-disclosure in the unraveling benchmark. The optimism encourages the sender to withhold any private information below the receiver's expectation of non-disclosure. For any given $\lambda_\rho \in (0, 1)$, the higher the public signal, the greater the optimism, and the stronger the incentive of strategic withholding.

Corollary 3. *For any $\theta_b > \theta_a^{min}$, holding θ_b fixed, $\theta_a^*(\theta_b)$ is strictly increasing in λ_ρ and ρ .*

A higher ρ (and thus higher λ_ρ) shifts more weight onto the public signal in the receiver's guess upon non-disclosure. For public signals that are above the minimum level, this shifts her belief more above the pessimism she would have held about non-disclosure in the unraveling benchmark, which in turn motivates the sender to increase the disclosure threshold in equilibrium.

Corollary 4. *We can decompose*

$$\theta_a^*(\theta_b) - \theta_a^{min} = \underbrace{\frac{2\lambda_\rho}{1 + \lambda_\rho} (\bar{\theta}_a - \theta_a^{min})}_{\text{baseline withholding due to failure in classical unraveling}} + \underbrace{\frac{2\lambda_\rho}{1 + \lambda_\rho} (\theta_b - \bar{\theta}_a)}_{\text{signal-contingent withholding}} \quad (4)$$

where $\bar{\theta}_a$ denotes the unconditional mean of $\mathcal{A}$.

Corollary 4 shows that the departure from full unraveling can be decomposed into two additive components: failure in classical unraveling and *signal-contingent withholding* (hereinafter *contingent withholding*). The classical unraveling failure component is present for any $\lambda_\rho > 0$ and is independent of θ_b . It corresponds to the baseline failure of full unraveling documented in Jin, Luca and Martin (2021), in which receivers' guesses upon non-disclosure are pulled toward the prior mean.⁸ It arises because, upon non-disclosure, receivers form beliefs as a convex combination of the average private state in the non-disclosure pool and the unconditional mean $\bar{\theta}_a$. As λ_ρ increases, more weight is placed on $\bar{\theta}_a$, leading to a larger departure from full unraveling.

⁷The proofs are in Appendix B.

⁸Jin, Luca and Martin (2021) can be viewed as a special case of our model ($\theta_b = \bar{\theta}_a$ and $\rho = 0$), in which the public signal coincides with the prior mean.

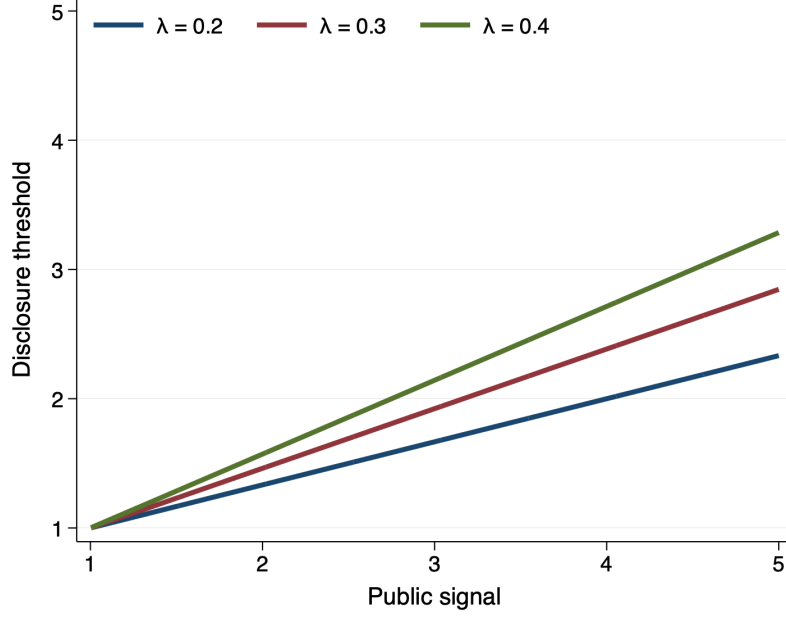


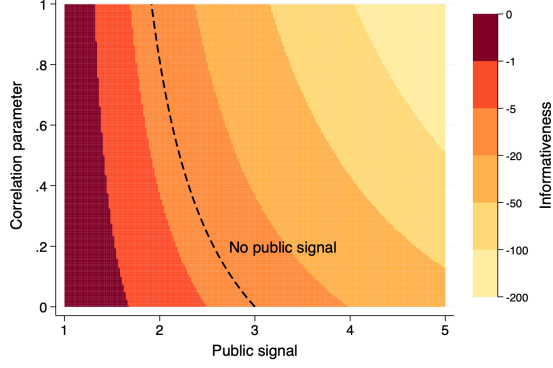
Figure 1: Simulated disclosure threshold

Note: This figure shows how the equilibrium disclosure thresholds change with the public signal. In this example, we set $\mathcal{B} = \mathcal{A} = [1, 5]$ and three levels of λ_ρ : 0.2, 0.3, and 0.4.

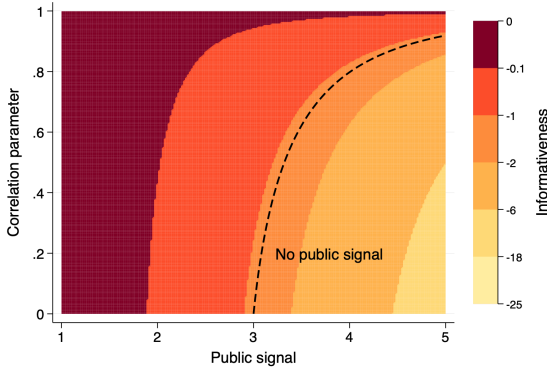
The contingent withholding component captures the additional effect of the public signal. When $\theta_b > \bar{\theta}_a$, this term is increasing in λ_ρ and ρ (assuming that λ_ρ increases in ρ), generating information crowd-out. When $\theta_b = \bar{\theta}_a$, the public signal coincides with the prior mean and provides no additional information, so the contingent withholding component vanishes. When $\theta_b < \bar{\theta}_a$, the contingent withholding component is negative, which generates information crowd-in: a sufficiently unfavorable public signal might offset the baseline failure in classical unraveling, reducing the departure from full unraveling.

Intuitively, when the public signal conveys good news ($\theta_b > \bar{\theta}_a$), the receiver places λ_ρ weight on a better-than-average prior upon non-disclosure; this optimism encourages the sender to withhold above a higher disclosure threshold, driving sender behavior further away from full unraveling. When the public signal conveys bad news ($\theta_b < \bar{\theta}_a$), the receiver places λ_ρ weight on a worse-than-average prior upon non-disclosure; this pessimism forces the sender to disclose above a lower disclosure threshold, bringing sender behavior closer to full unraveling. Information crowd-out and crowd-in are therefore two sides of the same mechanism: the receiver's weight on the public signal amplifies the informational content of θ_b into the non-disclosure inference in both directions.

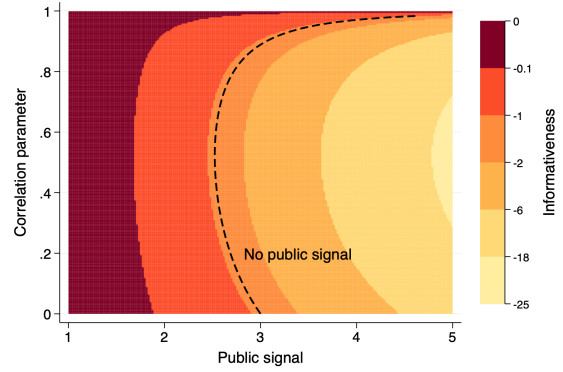
To illustrate the model predictions, in Figure 1 we plot how the equilibrium disclosure threshold changes with the public signal and receiver naivety. In this example, we set $\mathcal{B} = \mathcal{A} = [1, 5]$. The figure illustrates both the information crowd-out and the comparative statics in λ_ρ , showing that disclosure threshold is strictly increasing in the public signal θ_b , and strictly increasing in λ_ρ .



(a) Inference Accuracy (Non-Disclosure Environment)



(b) Informativeness (Full Environment), Case 1



(c) Informativeness (Full Environment), Case 2

Figure 2: Simulated inference accuracy and informativeness

Note: This figure shows how inference accuracy and informativeness change with the public signal and the correlation parameter. In this example, we set $\mathcal{B} = \mathcal{A} = [1, 5]$, and we define inference accuracy as $\mathcal{Q}(\theta_b, \lambda_\rho) = -100 \cdot \mathbb{E}[(\hat{\theta}_a - \theta_a)^2 | \theta_b, m = \emptyset]$ and informativeness as $\mathcal{I}(\theta_b, \lambda_\rho) = -100 \cdot \mathbb{E}[(\hat{\theta}_a - \theta_a)^2 | \theta_b]$. Panel (a) shows inference accuracy under the non-disclosure environment. Panels (b) and (c) show informativeness under the full environment (combining disclosure and non-disclosure). In Panels (a) and (c), we set $\lambda_\rho = 0.15 + 0.25\rho$; in Panel (b), we set $\lambda_\rho = 0.15 + 0.03\rho$. The dashed curve is the iso-informativeness curve that passes through $(3, 0)$, which represents the level of inference accuracy or informativeness under an environment without a public signal.

for any $\theta_b > 1$.

3.3 Inference Accuracy and Informativeness

Finally, we discuss the implications of our model for the information environment. We define inference accuracy as $\mathcal{Q}(\theta_b, \lambda_\rho) = -\mathbb{E}[(\hat{\theta}_a - \theta_a)^2 | \theta_b, m = \emptyset]$, the negative mean squared error of the receiver's guess conditional on non-disclosure, and informativeness as $\mathcal{I}(\theta_b, \lambda_\rho) = -\mathbb{E}[(\hat{\theta}_a - \theta_a)^2 | \theta_b]$, the same object computed over all realizations of the message. Both correspond to expected receiver payoffs under a quadratic loss function.⁹ Inference accuracy and informative-

⁹Similar definitions are used in Fr chet te, Lizzeri and Perego (2022), Farina et al. (2024) and Farina and Herman (2026). We adopt mean squared error rather than the 1.4-exponent loss used in our experimental design and empirical

ness correspond to two distinct environments. $\mathcal{Q}$ describes the non-disclosure environment and isolates receiver belief formation, holding fixed how often silence arises. $\mathcal{I}$ describes the full environment and therefore captures total information transmission, since it also reflects the frequency of disclosure. Because disclosure is verifiable, the receivers capture all the information transmitted when the senders disclose, so informativeness equals inference accuracy scaled by the probability of non-disclosure.

Given the equilibrium solution, we can derive $\mathcal{Q}(\theta_b, \lambda_\rho) = -\frac{4}{3} \cdot \frac{\lambda_\rho^2 (\theta_b - 1)^2}{(1 + \lambda_\rho)^2}$ and $\mathcal{I}(\theta_b, \lambda_\rho) = -\frac{8(1-\rho)}{3(\theta_a^{max}-1)} \cdot \frac{\lambda_\rho^3 (\theta_b - 1)^3}{(1 + \lambda_\rho)^3}$.¹⁰ Figure 2 plots simulated equilibrium values by public signal θ_b and correlation parameter ρ , with darker color indicating higher values. Panel (a) displays inference accuracy in the non-disclosure environment, and Panels (b) and (c) display informativeness under two parameterizations of the full environment. More specifically, in Panels (a) and (c), we set $\lambda_\rho = 0.15 + 0.25\rho$; in Panel (b), we set $\lambda_\rho = 0.15 + 0.03\rho$.

Equilibrium inference accuracy under non-disclosure is monotonically decreasing in both θ_b and λ_ρ , reaching its maximum of zero at $\theta_b = 1$ and $\lambda_\rho = 0$, the two cases in which the equilibrium exhibits full unraveling. A larger θ_b or λ_ρ widens the set of private states that are concealed and shifts the receiver's guess away from the rational benchmark toward the public signal itself. Both forces enlarge the gap between the receiver's guess and the realized state. This is illustrated in Panel (a) of Figure 2. Upon non-disclosure, inference accuracy is highest in the lower-left region and lowest in the upper-right.

Under the full environment, in which the disclosure probability also enters the measure, the monotonicity in ρ does not always hold. Two opposing forces determine the relationship between $\mathcal{I}$ and ρ . First, a higher ρ raises λ_ρ , widening the set of private states that are concealed and making disclosure less likely, conditional on a private state. Second, a higher ρ makes the private state coincide with the public signal more often; because our model predicts that the disclosure threshold is always lower than the public signal itself, this force makes disclosure weakly more likely. Disclosures contribute zero loss to informativeness, so the second force improves informativeness while the first worsens it. The net effect depends on the relative magnitude of the two forces and is governed by how strongly λ_ρ responds to ρ : when naivety responds weakly to the correlation parameter, the second force dominates and informativeness increases monotonically in ρ . In contrast, when λ_ρ responds to ρ sharply, the first force dominates over part of the range and informativeness becomes non-monotone in ρ .¹¹ We illustrate the two cases in Panel (b), where λ_ρ responds weakly to ρ , and Panel (c), where λ_ρ responds relatively sharply to ρ .

The dashed curves in Figure 2 are the iso-informativeness curve representing the level of inference accuracy or informativeness achievable in an environment without any public signal. The no-public-signal benchmark corresponds to $\rho = 0$ and $\lambda_0 = 0.15$, and the dashed curve passes

analysis because it admits a clean closed-form expression and simplifies the illustration of the comparative statics. The qualitative conclusions are identical under either definition.

¹⁰The non-disclosure probability is given by $(1 - \rho) \cdot \frac{\theta_a^*(\theta_b) - 1}{\theta_a^{max} - 1} = \frac{2(1-\rho)}{\theta_a^{max} - 1} \cdot \frac{\lambda_\rho (\theta_b - 1)}{1 + \lambda_\rho}$.

¹¹Note that, as $\rho \rightarrow 1$ (the public signal perfectly identifies the private state), $\mathcal{I} \rightarrow 0$.

through $(\bar{\theta}_a, 0) = (3, 0)$ in each panel.¹² Whenever (θ_b, ρ) falls to the right of the dashed curve, the public signal reduces inference accuracy or informativeness below the no-signal benchmark, and the crowd-out of voluntary disclosure dominates the direct informational contribution of the signal.

The region in which the public signal reduces inference accuracy or informativeness below the no-signal benchmark differs systematically across the three panels. In the non-disclosure environment, inference accuracy is decreasing in both θ_b and ρ , so the backfire region is large and expands along both dimensions. In the full environment, which combines disclosure and non-disclosure outcomes, the backfire region is substantially smaller and can be confined to intermediate values of the correlation parameter. Thus, public signals are most likely to backfire when θ_b is far from the lower bound of the state space and when correlation is high enough to distort receiver beliefs but not yet high enough to induce near-universal disclosure.

4 Experiment Design

In our experiments, subjects completed 50 rounds of the disclosure game. Instructions for the experiment were presented and read to the subjects at the beginning of the session. The Appendix contains the full instructions. At the end of each session, subjects were paid privately and in cash, receiving their show-up fee plus any additional earnings from the experiment. During the experiment, subjects earned their payoffs in “Experimental Currency units” (or ECUs). Two random rounds from each session were selected for payment. Each subject’s total ECUs in those two rounds were converted into U.S. dollars at a rate specified in the instructions, and her final payment was rounded to the nearest whole dollar amount.

Each round. In each round, subjects were randomly and anonymously paired. Subjects were informed that consecutive pairings with the same partner were unlikely. In each round and for each pairing, one subject was the sender and the other subject was the receiver. The sender was referred to as the “S Player,” and the receiver was referred to as the “R Player.”

For each pair, the computer draws a public signal $\theta_b \in \{1, 3, 5\}$. Each of these numbers was equally likely to be drawn. Both senders and receivers were informed of this uniform distribution over the state space, and the realized value of θ_b was observable to both players. Then, the computer draws a private state $\theta_a \in \{1, 2, 3, 4, 5\}$ using the following rule: with probability ρ , the private state θ_a is equal to the public signal θ_b ; with probability $1 - \rho$, θ_a is drawn from a uniform distribution on the set $\{1, 2, 3, 4, 5\}$. This procedure, along with the correlation parameter ρ , was clearly explained to both players during the instruction period.¹³ The sender observed both θ_a and θ_b , whereas the receiver observed only θ_b .

¹²We assign the prior mean as the hypothetical public signal. In practice, λ_0 may deviate from 0.15. However, it is expected to be positive, because receivers are found to be insufficiently sophisticated even in environments without correlated public signals (Jin, Luca and Martin, 2021), and small, because the prior mean is independent of the private state.

¹³See Appendix D for the experimental instructions.

Table 1: Treatment Variations

Treatment	$\pi(\theta_a \theta_b)$	$\mathbb{E}[\theta_a \theta_b]$
Low-correlation (LC)	$0.2 \cdot \mathbb{1}[\theta_a = \theta_b] + 0.16$	$0.2 \theta_b + 2.4$
Medium-correlation (MC)	$0.5 \cdot \mathbb{1}[\theta_a = \theta_b] + 0.10$	$0.5 \theta_b + 1.5$
High-correlation (HC)	$0.8 \cdot \mathbb{1}[\theta_a = \theta_b] + 0.04$	$0.8 \theta_b + 0.6$

Each sender then chose whether to report her private state θ_a to the receiver, with the understanding that the receiver observed the public signal θ_b . If the sender decided to report θ_a , the receiver observed the message “The number I received was,” followed by the actual θ_a . If the sender decided not to report, the receiver observed the message “No message.” After observing either a reported number or no message, the receiver was asked to guess the state θ_a ; guesses could be any half unit between 1 and 5, so the guess could take any value in the set $\{1, 1.5, 2, 2.5, 3, 3.5, 4, 4.5, 5\}$. There were no time constraints on the decisions of either the sender or the receiver. The action spaces of both players were common knowledge.

The sender’s payoff in each round was $U_S(\hat{\theta}_a) = 110 - 20|5 - \hat{\theta}_a|^{1.4}$, where $\hat{\theta}_a$ is the receiver’s guess. This payoff depends only on $\hat{\theta}_a$ and is strictly increasing in it. The receiver’s payoff in each round was $U_R(\hat{\theta}_a, \theta_a) = 110 - 20|\hat{\theta}_a - \theta_a|^{1.4}$, which decreases strictly as the receiver’s guess moves further from the realized θ_a in either direction.¹⁴ The introduction of the public signal θ_b does not alter these payoff functions; it affects only the information available to both parties when the sender makes the disclosure decision. Because the state space and action space are small, these payoffs can be presented to subjects as a table, so that participants do not need to know or interpret the functional forms directly.

Feedback was provided to both players at the end of each round. Both the sender and the receiver were told the private state θ_a , the sender’s disclosure decision, the receiver’s guess of θ_a , and their own payoff in this round.

Treatment Variation. Our primary treatment variations occurred in the correlation between the private state θ_a and the public signal θ_b . We consider three treatments: low-correlation (LC), medium-correlation (MC), and high-correlation (HC). These correspond to values of the correlation parameter ρ of 0.2, 0.5, and 0.8, respectively. Table 1 summarizes the treatment variations, along with the corresponding conditional distribution and expectation of θ_a .

Belief Elicitation. After completing 50 rounds, subjects were asked to report their beliefs about (i) the frequency with which senders reported each possible private state θ_a , given each public signal θ_b , and (ii) the average receiver guess conditional on θ_a not being reported, given each public signal θ_b , during the preceding 50 rounds. These questions were designed to assess whether beliefs about sender strategies influenced receiver decisions, and whether beliefs about receiver behavior influenced sender decisions.¹⁵

¹⁴The payoff functions are identical to those in Jin, Luca and Martin (2021).

¹⁵There are two potential limitations of our belief elicitation tasks. First, they are not incentivized, which may introduce measurement error (Trautmann and van de Kuilen, 2015). Second, recovering the payoff-relevant beliefs for

Testable Hypotheses. We consider the non-trivial case in which $\lambda_\rho > 0$. In our setting, $\mathcal{A} = \{1, 2, 3, 4, 5\}$ and $\mathcal{B} = \{1, 3, 5\}$. Section 3 establishes that a unique equilibrium exists, with disclosure threshold

$$\theta_a^*(\theta_b) = \frac{1 - \lambda_\rho}{1 + \lambda_\rho} + \frac{2\lambda_\rho}{1 + \lambda_\rho} \theta_b \quad (5)$$

To link the equilibrium predictions to testable hypotheses, we translate changes in disclosure thresholds into changes in disclosure rates. Formally, we define

$$\begin{aligned} \theta_a^*(\theta_{b_1}, \rho_1) > \theta_a^*(\theta_{b_2}, \rho_2) &\iff \Pr(d = 1 | \theta_a, \theta_{b_1}, \rho_1) \leq \Pr(d = 1 | \theta_a, \theta_{b_2}, \rho_2) \quad \forall \theta_a \\ &\text{and } \exists \theta_a \text{ such that } \Pr(d = 1 | \theta_a, \theta_{b_1}, \rho_1) < \Pr(d = 1 | \theta_a, \theta_{b_2}, \rho_2) \end{aligned} \quad (6)$$

In other words, the disclosure threshold under (θ_{b_1}, ρ_1) exceeds that under (θ_{b_2}, ρ_2) if and only if, for every θ_a , the disclosure rate under (θ_{b_1}, ρ_1) is weakly lower than that under (θ_{b_2}, ρ_2) , with strict inequality for at least one θ_a .

While Equation 5 is derived on a continuum, in Appendix C we characterize the equilibrium of the discrete game induced by our design and verify that the hypotheses below hold in it, with every equilibrium cutoff lying within $2/(1 + \lambda_\rho)$ of the threshold in Equation 5.

Now, we explain how we translate the four corollaries of Section 3 into two main testable hypotheses. The first is information crowd-out: Corollary 2 predicts that the disclosure threshold increases in θ_b . Equivalently, the disclosure rate shall decrease in θ_b . Information crowd-out arises because a more favorable public signal raises receiver beliefs upon non-disclosure, which discourages disclosure and enlarges the set of types that conceal in equilibrium.

Hypothesis 1. *[Better public signal generates more information crowd-out in disclosure.] Holding the private state θ_a fixed, sender disclosure rates are decreasing in the public signal θ_b , while receiver beliefs following non-disclosure are increasing in θ_b .*

The second prediction concerns how the equilibrium threshold varies with the correlation parameter ρ . Assuming that λ_ρ increases in ρ , Corollary 3 implies that $\theta_a^*(\theta_b)$ is weakly increasing in ρ for every θ_b , and strictly increasing in ρ when $\theta_b > 1$.

When the public signal is at the lower bound of the state space ($\theta_b = 1$), the threshold equals $\theta_a^*(1) = 1$, regardless of ρ . The public signal coincides with the minimum possible state, leaving no scope for the receiver's belief following non-disclosure to be pulled upward, so disclosure incentives are unaffected by ρ . When $\theta_b \in \{3, 5\}$, a higher ρ increases the probability that the private state equals the public signal, and this raises the receiver's belief following non-disclosure, weakens the sender's incentive to disclose, and increases the equilibrium threshold. The effect is larger at $\theta_b = 5$ than at $\theta_b = 3$, since the public signal pulls receiver beliefs further above the non-disclosure pool mean.

receivers requires imposing Bayesian updating on elicited beliefs, a potentially strong assumption given documented deviations from rationality in sender-receiver games (Farina and Leccese, 2024). Our belief data are intended as suggestive evidence on the channels behind disclosure crowd-out and receiver naivety. We also show that our results are robust to restricting attention to the final 20 rounds, over which elicited beliefs most plausibly reflect subject decisions.

Hypothesis 2. *[Crowd-out from public signal only occurs when the public signal is above minimum and it is exacerbated by signal correlation.] Disclosure rates do not vary with ρ when $\theta_b = 1$, so that $LC = MC = HC$. When $\theta_b \in \{3, 5\}$, disclosure rates are strictly decreasing in ρ , so that $LC > MC > HC$. Receiver beliefs following non-disclosure are correspondingly invariant to ρ at $\theta_b = 1$ and strictly increasing in ρ at $\theta_b \in \{3, 5\}$.*

5 Results

This section presents the main empirical findings. We first document how sender disclosure rates and receiver guesses respond to the public signal and to the correlation between the public signal and the private state, confirming Hypotheses 1 and 2. We then examine senders' and receivers' stated beliefs about each other, which point to receiver naivety as the central deviation from rational inference. Next, we estimate the receiver naivety parameter in our model, confirm that it increases monotonically with the correlation between the private state and the public signal, and show that the model fits the data more closely than alternative specifications. Finally, we establish the implications of these behaviors for information transmission, showing that the public signal can backfire when receiver naivety is high and the public signal is favorable.

Our main sessions were conducted at the Experimental Economics Lab at the University of Maryland (EEL-UMD), with IRB approval from UMD. In this laboratory, subjects were separated by dividers, and each subject was provided with a personal computer terminal. Our experiment was programmed using oTree (Chen, Schonger and Wickens, 2016). The full experimental instructions and screenshots are provided in Appendix D and E.

We assigned 66 subjects to the low-correlation (LC) treatment, 62 subjects to the medium-correlation (MC) treatment, and 110 subjects to the high-correlation (HC) treatment.¹⁶ Sessions were randomly assigned to one of the three treatments, and all subjects within a session faced the same value of ρ . None of the 238 subjects participated in more than one session.¹⁷ Roles were randomly reassigned at the start of each round, so that every subject played as both sender and receiver over the course of the experiment. We used a show-up fee of \$10, and on average subjects earned \$19.85. The minimum payment was \$10, and the maximum payment was \$24.

5.1 Sender Disclosures and Receiver Guesses

We begin by documenting how the sender disclosure rates change by the public signal. In Figure 3, we pool senders' decisions across rounds and treatments, and present the disclosure rates by

¹⁶This corresponds to 5, 4, and 8 sessions for LC, MC, and HC, respectively. By construction, the HC treatment generates fewer observations where the private state differs from the public signal, so we recruited additional subjects in HC to maintain sample balance across (θ_a, θ_b) pairs.

¹⁷In terms of demographics, we had roughly an equal number of men and women. The majority of subjects identified as White or Asian (82 percent), and the majority reported English as their native language (89 percent). Less than 9 percent of subjects reported having a friend present in the room during the session. Appendix Table A1 reports the full summary statistics. In the regressions presented in this section, we either control for all demographic characteristics or include subject fixed effects.

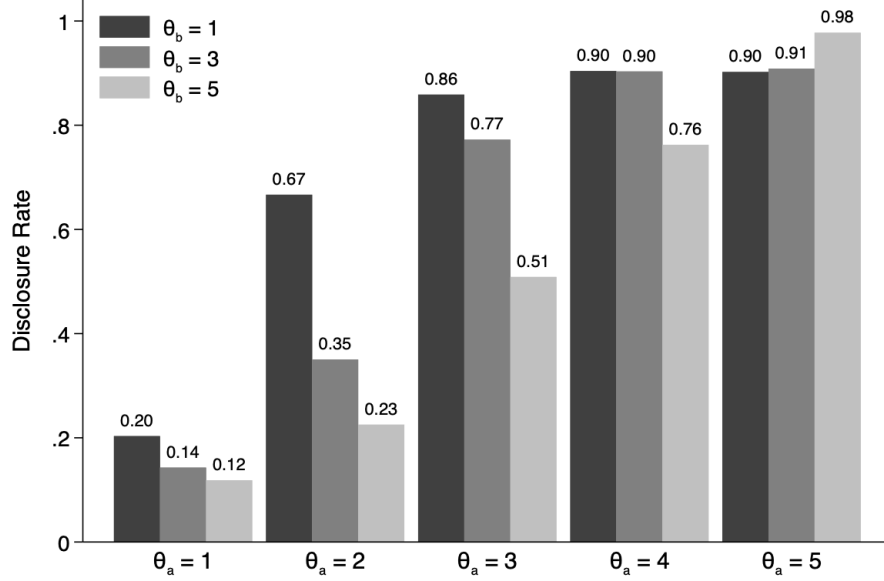


Figure 3: Sender Disclosure Rates by Public Signal

the private state and the public signal.

There are two main takeaways from this figure. First, disclosure rates increase monotonically in the private state θ_a : conditional on any realization of the public signal, senders with more favorable private information are substantially more likely to report it, rising from 0.12–0.20 at $\theta_a = 1$ to 0.90–0.98 at $\theta_a = 5$. This reflects the incentive to separate from weaker types through disclosure.

Second, holding the private state (θ_a) fixed, disclosure rates tend to decline in the public signal θ_b , with the decline most pronounced for intermediate values of the private state. When $\theta_a = 2$, disclosure drops from 0.67 when $\theta_b = 1$ to 0.35 when $\theta_b = 3$ and to 0.23 when $\theta_b = 5$ (p -value < 0.01 for all pairwise comparisons), a decline of roughly 44 percentage points across the range of θ_b . A similar pattern arises when $\theta_a = 3$, where disclosure falls from 0.86 at $\theta_b = 1$ to 0.51 at $\theta_b = 5$ (p -value < 0.01 for all pairwise comparisons). When $\theta_a = 1$ and $\theta_a = 4$, the disclosure rates move in the same direction, though the changes in magnitude are smaller and are not always statistically significant.¹⁸ When $\theta_a = 5$, senders with $\theta_b = 5$ are more likely to disclose than the others, but over 90 percent of senders disclose across all values of the public signal, leaving little room for variation.

Therefore, these results establish information crowd-out: voluntary disclosure declines as the public signal becomes more favorable, with the effect concentrated at intermediate values of the private state, consistent with Hypothesis 1.

¹⁸At $\theta_a = 1$, p -value < 0.01 for $\theta_b = 1$ vs. $\theta_b = 5$, p -value = 0.06 for $\theta_b = 1$ vs. $\theta_b = 3$, and p -value = 0.49 for $\theta_b = 3$ vs. $\theta_b = 5$; at $\theta_a = 4$, p -value < 0.01 for $\theta_b = 1$ vs. $\theta_b = 5$ and $\theta_b = 3$ vs. $\theta_b = 5$, and p -value = 0.98 for $\theta_b = 1$ vs. $\theta_b = 3$.

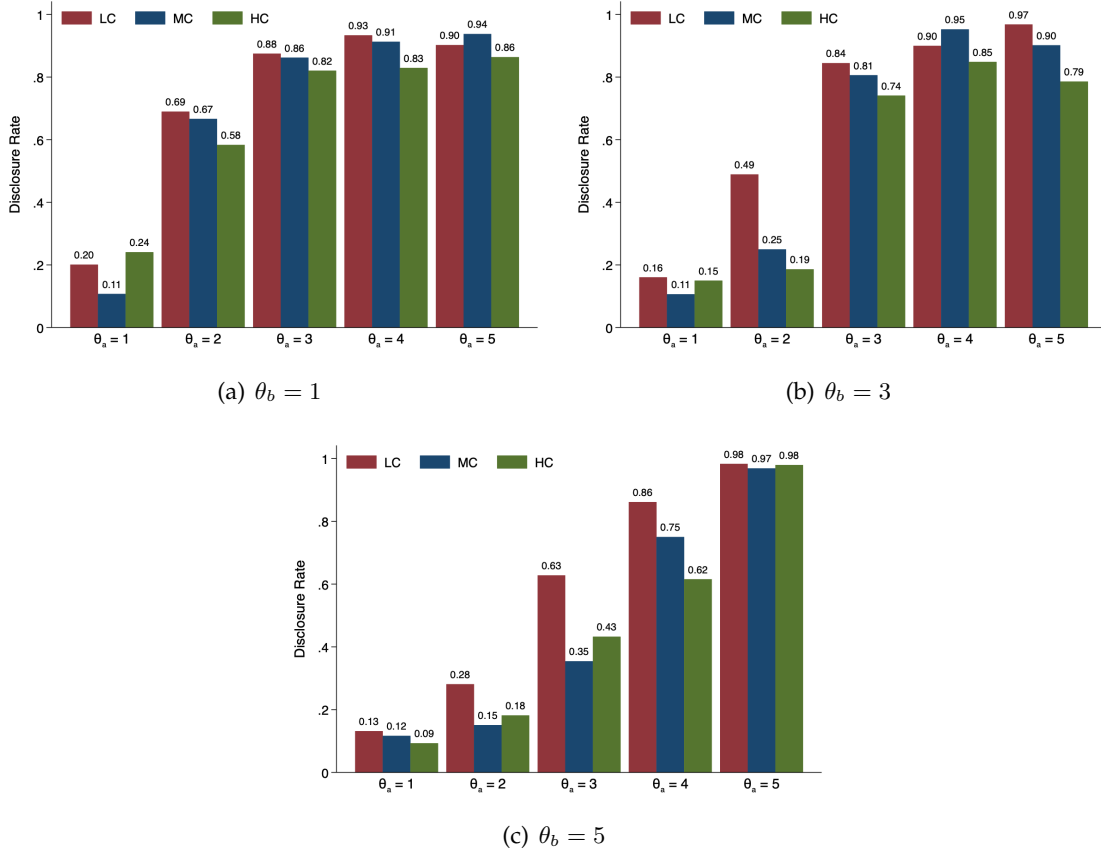


Figure 4: Sender Disclosure Rates by Treatment

Result 1. Holding the private state fixed, sender disclosure rates (weakly) decline in the public signal. The decline is largest at intermediate values of the private state, and is attenuated at the extremes of the state space.

Next, we consider how senders' disclosure decisions vary by the correlation between the public signal and the private state. Panel (a) of Figure 4 shows disclosure rates at $\theta_b = 1$. Across all five values of θ_a , disclosure rates display no clear monotone ordering in ρ . When $\theta_a = 1$, the HC rate (0.24) slightly exceeds the LC rate (0.20), with MC below both (0.11); but all three values are low, and at $\theta_a = 1$ senders are weakly better off concealing regardless of treatment. When $\theta_a \in \{2, 3, 4, 5\}$, disclosure rates vary little across treatments, with p -values exceeding 0.1 for all pairwise comparisons but one. Intuitively, when $\theta_b = 1$ coincides with the lowest possible state, a more predictive signal cannot pull receiver beliefs any lower, leaving disclosure incentives invariant to ρ .

Panels (b) and (c) of Figure 4 report the disclosure rates at $\theta_b = 3$ and $\theta_b = 5$, respectively. In both panels, disclosure rates decline in the correlation between θ_a and θ_b for intermediate values of the private state. At $\theta_b = 3$, the decline is most pronounced when $\theta_a = 2$: the disclosure rate falls from 0.49 in LC to 0.25 in MC and to 0.19 in HC, a 30-percentage-point drop across the range

of ρ . A weaker but similar pattern arises at $\theta_a = 3$, where the rate falls from 0.84 in *LC* to 0.74 in *HC*. At $\theta_b = 5$, the decline is observed at $\theta_a \in \{2, 3, 4\}$, when the disclosure rates drop from 0.28, 0.63, and 0.86 in *LC* to 0.18, 0.43, and 0.62 in *HC*, respectively.¹⁹

These patterns are consistent with Hypothesis 2. At $\theta_b = 1$, disclosure rates show no systematic pattern across treatments, and senders' behavior approaches full unraveling: senders disclose at rates above 0.82 for all $\theta_a \geq 3$, and above 0.58 at $\theta_a = 2$ regardless of treatment. At $\theta_b \in \{3, 5\}$, a stronger correlation generates stronger crowd-out, with disclosure rates lowest under *HC* and highest under *LC*. Similar to the information crowd-out observed in Figure 3, the treatment effect is most pronounced at intermediate values of the private state.

Result 2. *When the public signal is low, sender disclosure is invariant to the correlation between the private state and the public signal. When the public signal is intermediate or high, higher correlation reduces disclosure, with the largest decline at intermediate private states.*

We now discuss the receivers' guesses. Table 2 reports the average receiver guess conditional on each sender report. The table also reports the average guess upon non-disclosure, along with its deviation from the private state, $\hat{\theta}_a - \theta_a$.

First, when senders disclose, receivers' guesses are largely accurate, with the smallest errors at $\theta_a = 3$. Receivers tend to overguess at the low extreme ($\theta_a \in \{1, 2\}$) and underguess at the high extreme ($\theta_a \in \{4, 5\}$).²⁰ The deviations are largest at $\theta_a = 1$, where the average guess ranges from 1.00 to 1.64, and at $\theta_a = 4$, where it ranges from 3.59 to 4.08. Receivers nonetheless treat disclosed reports as credible, consistent with the verifiable nature of the disclosure game.

Second, receiver guesses following non-disclosure vary systematically with both the public signal and the correlation between the public signal and the private state. Holding the treatment fixed, guesses following non-disclosure increase in θ_b : in *LC*, the average non-disclosure guess increases from 1.39 at $\theta_b = 1$ to 2.28 at $\theta_b = 3$ and to 2.41 at $\theta_b = 5$, and the same qualitative ordering holds in *MC* and *HC*. Holding the public signal fixed, guesses following non-disclosure increase also in the correlation between θ_a and θ_b for $\theta_b \in \{3, 5\}$. At $\theta_b = 3$, the non-disclosure guess increases from 2.28 in *LC* to 2.41 in *MC* and to 2.70 in *HC*. At $\theta_b = 5$, the corresponding sequence is 2.41, 3.02, and 3.39.²¹ These patterns mirror the sender-side evidence on information crowd-out: receiver guesses following non-disclosure increase with the public signal, and this increase is amplified by a stronger correlation between the private state and the public signal, particularly when $\theta_b \in \{3, 5\}$.

At $\theta_b = 1$, the equilibrium predicts a non-disclosure guess of exactly one regardless of ρ ; in the data, guesses are slightly above one under *LC* (1.39) and decline toward the prediction as ρ rises (1.24 in *MC* and 1.15 in *HC*).

¹⁹Panel A of Table A2 reports the p -values of all pairwise tests of the treatment effects in sender disclosure rates. The vast majority of treatment differences are statistically significant, with p -value < 0.01 .

²⁰This pattern is also observed in Jin, Luca and Martin (2021).

²¹Panel B of Table A2 reports the p -values of all pairwise tests of the treatment effects in receiver guesses upon non-disclosure. All except one treatment difference has p -value < 0.05 .

Table 2: Summary of Receiver Guesses

	LC		MC		HC	
	N	Guess	N	Guess	N	Guess
When $\theta_b = 1$						
Guess (report=1)	80	1.363	64	1.125	374	1.273
Guess (report=2)	120	2.150	68	1.985	28	2.000
Guess (report=3)	140	3.050	100	3.000	64	2.781
Guess (report=4)	168	3.964	84	3.714	68	3.588
Guess (report=5)	166	4.687	90	4.600	76	4.776
Guess (no report)	211	1.389	298	1.235	620	1.147
Guess – θ_a (no report)	211	-0.090	298	0.044	620	0.035
When $\theta_b = 3$						
Guess (report=1)	26	1.192	10	1.000	12	1.333
Guess (report=2)	92	2.152	30	2.100	16	2.250
Guess (report=3)	304	3.102	490	3.033	1098	3.059
Guess (report=4)	144	4.069	80	4.013	56	3.893
Guess (report=5)	182	4.901	110	4.964	66	4.818
Guess (no report)	155	2.277	154	2.406	275	2.698
Guess – θ_a (no report)	155	0.374	154	0.153	275	-0.011
When $\theta_b = 5$						
Guess (report=1)	24	1.625	14	1.643	8	1.000
Guess (report=2)	54	2.222	16	2.188	12	2.750
Guess (report=3)	108	3.111	34	3.118	32	3.094
Guess (report=4)	112	4.080	84	4.012	48	4.042
Guess (report=5)	462	4.885	616	4.917	1516	4.954
Guess (no report)	193	2.407	153	3.023	118	3.386
Guess – θ_a (no report)	193	0.495	153	0.788	118	0.877

Turning to the deviation of the receiver's guess from the true private state upon non-disclosure: at $\theta_b = 1$, the deviation is close to zero across all three treatments, ranging from -0.09 in *LC* to 0.04 in *MC* and 0.04 in *HC*, indicating that receivers are correct on average when the public signal is low. At $\theta_b = 3$, the deviation is positive in *LC* (0.37) and *MC* (0.15), and close to zero in *HC* (-0.01); the deviation shrinks as the correlation between θ_a and θ_b grows, as both the receiver's guess and the withheld private states converge toward 3.²² At $\theta_b = 5$, the deviation is uniformly positive and increases in the correlation between θ_a and θ_b , rising from 0.50 in *LC* to 0.79 in *MC* and 0.88 in *HC*. Intuitively, at $\theta_b = 5$, a receiver that leans on the public signal overguesses the true state,

²²Our model predicts that the average overguess is $\frac{\lambda_\rho}{1+\lambda_\rho} \cdot (\theta_b - \theta_a^{min})$, which should increase with λ_ρ whenever $\theta_b > \theta_a^{min}$. Here we find an opposing trend (although the only pair that is statistically significantly different is 0.374 vs. -0.011) mainly due to the empirical differences in disclosure rates at high private states, which reflect subject heterogeneity. In a hypothetical environment in which disclosure rates are the same across treatments at $\theta_a \geq 3$ (e.g., always disclose), the overguess increases in ρ at $\theta_b = 3$.

Table 3: Regressions on Sender Disclosures and Receiver Guesses

	Sender disclosures (0/1)		Receiver guesses (no report)	
	(1)	(2)	(3)	(4)
MC	-0.042*		0.201*	
	(0.024)		(0.099)	
HC	-0.029		0.251**	
	(0.026)		(0.093)	
θ_a	0.216***	0.215***		
	(0.007)	(0.004)		
θ_b	-0.034***	-0.032***	0.455***	0.466***
	(0.005)	(0.004)	(0.044)	(0.046)
Demographics	X		X	
Subject FE		X		X
Observations	5,950	5,950	2,177	2,177

Notes: Columns (1) and (2) report OLS estimates of the sender's disclosure decision, an indicator equal to one when the sender reports θ_a . Columns (3) and (4) report OLS estimates of the receiver's guess, restricted to rounds in which the sender did not report. LC is the omitted treatment category. Demographic controls include age, gender, race, major, class year, native language, presence of a friend in the session, and elicited risk preference. Standard errors, clustered at the session level, are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

and the overguess widens as the correlation between θ_a and θ_b grows because the receiver places more weight on the public signal.

Result 3. *Following non-disclosure, (i) receiver guesses increase in the public signal, (ii) holding the public signal fixed, receiver guesses increase in the correlation between θ_a and θ_b when the public signal is intermediate or high, and (iii) receivers tend to overguess the private state, and the overguess is largest in high public signal and high correlation environments.*

These findings are robust in regressions that control for subject demographics or subject fixed effects. Table 3 reports regression estimates for sender disclosure decisions and receiver guesses upon non-disclosure. Standard errors are clustered at the session level. Disclosure rates increase significantly in θ_a and fall in θ_b , while non-disclosure guesses increase in θ_b . Moreover, disclosure rates are lower and receiver guesses correspondingly higher under MC and HC than under LC.²³ These estimates are stable across different specifications.

Finally, we discuss sender and receiver learning during the experiment. Figure A.1 plots sender disclosure rates by round. The most visible round-by-round movement occurs at intermediate private states. At medium θ_a ($\theta_a \in \{2, 3\}$), disclosure rates increase in every treatment from the early rounds to the late rounds, climbing from around 0.6 to close to 0.8 by round 50. This is also the region of the state space where we expect learning to be most visible, because disclosure rates exhibit the most variation across public signals and treatments, and sender decisions at

²³While the coefficient on MC is higher than HC in the first column, they are not statistically significantly different.

intermediate θ_a are most sensitive to receiver guesses. At low θ_a , disclosure rates remain roughly flat across treatments and rounds, as senders are weakly better off remaining silent regardless of receiver beliefs. At high θ_a , disclosure rates are close to one throughout and display no meaningful trend.²⁴

Table A4 reports the regression results of the change in the receiver’s non-disclosure guess on indicators for whether she overguessed or underguessed the private state in the previous round. The results suggest that, when the public signal is either low ($\theta_b = 1$) or high ($\theta_b = 5$), receivers who overguessed the private state lower their subsequent guess upon non-disclosure, and the adjustment is much larger in *HC*. The change in receiver guesses at medium public signal ($\theta_b = 3$) is small and insignificant. By contrast, the effect of having underguessed is small and mostly insignificant.²⁵

In the Appendix Figure A.2 (sender disclosures by public signal), Figure A.3 (sender disclosures by treatment), Table A5 (summary of receiver guesses), Table A6 (regression results), we repeat the analysis above, restricting the sample to sender and receiver decisions in the final 20 rounds of the experiment. Our results are robust for this filtered subsample.

5.2 Stated Beliefs

We now examine senders’ and receivers’ stated beliefs about each other, elicited at the end of the experiment. Senders are largely accurate and best-respond to their beliefs; receivers are qualitatively correct but quantitatively off, with guesses that depart from the Bayesian benchmark implied by their stated beliefs. The combined evidence points to receiver naivety as the mechanism behind the disclosure crowd-out.

Table 4 compares senders’ stated beliefs to the realized average receiver guess, and reports the share of disclosure decisions that are consistent with the stated belief. A disclosure decision is defined as consistent with the stated belief if the sender discloses whenever the private state is weakly above her belief and remains silent whenever the private state is weakly below it.

Senders’ beliefs about receiver guesses are broadly accurate and match the comparative statics in the data. When the public signal is low, stated beliefs are close to one across all treatments, slightly above the realized non-disclosure guesses. When the signal is intermediate or high, senders correctly anticipate that receiver guesses increase with the public signal and with the correlation between the private state and the public signal, although they overestimate the receiver guesses by 0.24–0.52 when $\theta_b = 5$.

Moreover, sender disclosure decisions are largely consistent with stated beliefs. Across all treatments and signal realizations, between 84 and 96 percent of decisions align with the sender’s reported belief. Senders therefore appear to best-respond to their stated beliefs, and the informa-

²⁴In Table A3, we report regression estimates that confirm these results: sender disclosure rates increase over time; correspondingly, receiver guesses decrease over the course of the experiment.

²⁵This is consistent with the findings of Jin, Luca and Martin (2021), who report a similar asymmetry between overguessing and underguessing adjustments.

Table 4: Accuracy and Consistency of Sender Beliefs and Behavior

	LC	MC	HC
When $\theta_b = 1$			
Belief of receiver guess (no report)	1.52	1.29	1.33
Average receiver guess (no report)	1.39	1.23	1.15
Action consistent with self-reported belief	0.85	0.92	0.91
Rounds 1-15	0.81	0.88	0.90
Rounds 16-30	0.85	0.95	0.92
Rounds 31-50	0.88	0.92	0.92
When $\theta_b = 3$			
Belief of receiver guess (no report)	2.06	2.48	2.54
Average receiver guess (no report)	2.28	2.41	2.70
Action consistent with self-reported belief	0.84	0.89	0.93
Rounds 1-15	0.85	0.83	0.89
Rounds 16-30	0.82	0.89	0.93
Rounds 31-50	0.86	0.92	0.96
When $\theta_b = 5$			
Belief of receiver guess (no report)	2.65	3.45	3.91
Average receiver guess (no report)	2.41	3.02	3.39
Action consistent with self-reported belief	0.86	0.90	0.96
Rounds 1-15	0.85	0.89	0.95
Rounds 16-30	0.88	0.89	0.96
Rounds 31-50	0.86	0.91	0.97

* Notes: In this table, we drop individuals whose stated sender's beliefs are obviously irrational (i.e. they fail to increase weakly in θ_b). This represents 10.9 percent of all subjects. A disclosure decision is defined as consistent with the stated belief if the sender discloses whenever the private state is weakly above her belief and remains silent whenever the private state is weakly below it.

tion crowd-out reflects strategic responses rather than noise.²⁶ The consistency between disclosure decisions and stated beliefs improves over the course of the experiment. For example, at $\theta_b = 3$ in the HC treatment, consistency increases from 0.89 in rounds 1–15 to 0.96 in rounds 31–50, with similar patterns across treatments. This increase is partly mechanical, as beliefs are elicited at the end of the experiment and therefore align more closely with late-round behavior.

Receivers' stated beliefs reveal a different pattern. Table A7 summarizes receivers' stated beliefs about sender reporting rates at each combination of θ_b and θ_a . Receivers correctly per-

²⁶Regressions of disclosure decisions on stated beliefs support this pattern (Table A3). Senders who believe the receiver's non-disclosure guess will fall below their private state are more likely to disclose, with the effect largest in MC, where this belief raises the disclosure probability by roughly 17 percentage points. The estimates are stable across different regression specifications.

Table 5: Accuracy and Consistency of Receiver Guesses of Nonreported Number

	LC	MC	HC
When $\theta_b = 1$			
Implied belief of nonreported θ_a	1.76	1.39	1.36
Guess of θ_a	1.39	1.23	1.15
Rounds 31-50	1.44	1.15	1.05
Actual θ_a	1.48	1.19	1.11
Rounds 31-50	1.19	1.15	1.08
Guess – θ_a	-0.09	0.04	0.04
Rounds 31-50	0.25	0.00	-0.03
When $\theta_b = 3$			
Implied belief of nonreported θ_a	2.22	2.31	2.68
Guess of θ_a	2.28	2.41	2.70
Rounds 31-50	2.26	2.19	2.65
Actual θ_a	1.90	2.25	2.71
Rounds 31-50	1.82	1.98	2.56
Guess – θ_a	0.37	0.15	-0.01
Rounds 31-50	0.44	0.21	0.09
When $\theta_b = 5$			
Implied belief of nonreported θ_a	2.15	2.31	2.70
Guess of θ_a	2.41	3.02	3.39
Rounds 31-50	2.06	2.68	3.27
Actual θ_a	1.91	2.24	2.51
Rounds 31-50	1.81	2.29	2.54
Guess – θ_a	0.49	0.79	0.88
Rounds 31-50	0.25	0.39	0.73

* Notes: The implied belief is the posterior mean of θ_a conditional on non-disclosure, obtained by applying Bayes' rule to the stated reporting rates in Table A7 and the treatment-specific conditional distribution of θ_a given θ_b .

ceive that (i) sender reporting rates increase in θ_a , (ii) decrease in θ_b holding θ_a fixed, and (iii) at $\theta_b \in \{3, 5\}$, decline in the correlation between θ_a and θ_b , particularly for intermediate values of θ_a . Receivers nonetheless underestimate the level of reporting rates: when $\theta_b \in \{1, 3\}$ and $\theta_a \in \{4, 5\}$, believed reporting rates sit between 0.67 and 0.85, while the realized rates often exceed 0.90.

Receivers' actual guesses depart from the Bayesian guess implied by their stated beliefs about sender behavior. Table 5 reports the Bayesian guess of θ_a implied by the stated beliefs, alongside two benchmarks: receivers' actual guess and the empirical average of the true private state conditional on non-disclosure.²⁷ When $\theta_b \in \{1, 3\}$, the implied guess is close to the actual guess, differing by at most 0.10 across all three treatments at $\theta_b = 3$. At $\theta_b = 5$, by contrast, the implied

²⁷The implied belief is computed using the reporting rates from Table A7 and the treatment-specific conditional distribution of θ_a given θ_b , and assumes receivers correctly invert their beliefs about sender behavior into a posterior over the private state.

guess falls substantially below the actual guess, and the gap widens with ρ : the implied guess is 2.15, 2.31, and 2.70 in *LC*, *MC*, and *HC*, while the actual guess is 2.41, 3.02, and 3.39.²⁸ Two interpretations are consistent with this gap: receivers' true beliefs about sender reporting rates may differ from those they state, or they may fail to fully Bayes-update from their stated beliefs to a posterior over the private state. Either interpretation motivates the receiver naivety model that we estimate in the next subsection.

Finally, receivers' behavior shifts modestly toward the implied belief in late rounds. Restricting to the final 20 rounds, the deviation of the actual guess from the realized private state at $\theta_b = 5$ falls from 0.49, 0.79, and 0.88 to 0.25, 0.39, and 0.73 across *LC*, *MC*, and *HC*, narrowing the gap to the implied belief; deviations at $\theta_b = 3$, where actual and implied guesses are already close, do not shrink. The pattern is consistent with partial learning at $\theta_b = 5$, where feedback is most informative, but a substantial gap between the actual guess and the implied belief persists even by the end of the experiment.

5.3 Estimation of Receiver Naivety and Model Fit

Building on the model introduced in Section 3, we now estimate the receiver naivety parameter. For each treatment, we restrict attention to receiver decisions following non-disclosure and model the guess of receiver i in round t as

$$g_{it} = (1 - \lambda_\rho) \mu_\rho(\theta_{b,it}) + \lambda_\rho \theta_{b,it} + \epsilon_{it} \quad (7)$$

where $\mu_\rho(\theta_{b,it})$ is the conditional expectation of θ_a given non-disclosure and θ_b , implied by the empirical sender strategy. Let $\hat{\sigma}_\rho(\theta_a, \theta_b)$ denote the empirical disclosure rate of senders observing (θ_a, θ_b) in treatment ρ , and $p_\rho(\theta_a | \theta_b)$ denote the prior on θ_a given θ_b implied by our design. Then Bayes' rule yields

$$\mu_\rho(\theta_b) = \sum_{a=1}^5 \theta_a \cdot \frac{p_\rho(\theta_a | \theta_b) [1 - \hat{\sigma}_\rho(\theta_a, \theta_b)]}{\sum_{a'=1}^5 p_\rho(\theta_{a'} | \theta_b) [1 - \hat{\sigma}_\rho(\theta_{a'}, \theta_b)]} \quad (8)$$

This is the guess of a sophisticated receiver who internalizes the strategic content of silence. The alternative reference point, θ_b , is the guess of a naive receiver who treats the public signal as a direct prediction of θ_a . The parameter λ_ρ measures the weight placed on the naive reference point and therefore captures receiver naivety in each treatment.

Defining $y_{it} = g_{it} - \mu_\rho(\theta_{b,it})$ and $x_{it} = \theta_{b,it} - \mu_\rho(\theta_{b,it})$, we estimate

$$y_{it} = \lambda_\rho x_{it} + \epsilon_{it} \quad (9)$$

using OLS separately for each treatment, clustering standard errors at the session level.

²⁸Regressions of receiver guesses on the implied Bayesian belief confirm that receivers respond to their own beliefs (Table A3). A one-unit increase in the implied belief is associated with a 0.36-unit increase in the actual guess, statistically significant at the one-percent level. The response is nonetheless short of the unit coefficient that full Bayesian updating would imply.

Table 6: Estimation of Receiver Naivety

	LC	MC	HC
λ_ρ (all rounds)	0.169*** (0.023)	0.262** (0.064)	0.320*** (0.059)
p -value (vs. LC)		0.145	0.025
Observations	559	605	1,013
λ_ρ (rounds 1-30)	0.219*** (0.028)	0.328** (0.093)	0.353*** (0.083)
p -value (vs. LC)		0.227	0.131
Observations	362	385	631
λ_ρ (rounds 31-50)	0.092 (0.060)	0.149* (0.054)	0.261** (0.095)
p -value (vs. LC)		0.454	0.134
Observations	197	220	382

Notes: Standard errors are clustered at the session level. Each p -value tests the equality of the treatment estimate and the LC estimate, based on a Wald test from a stacked regression that interacts the estimating equation with treatment indicators. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 6 reports estimates using all rounds, the first 30 rounds, and the last 20 rounds. Pooling all 50 rounds, $\hat{\lambda}_\rho$ is positive and statistically different from zero in every treatment, indicating that receivers place non-trivial weight on the public signal beyond the sophisticated benchmark. Receiver naivety is larger in the first 30 rounds than in the last 20 across all three treatments, consistent with learning over the course of the experiment. In the final 20 rounds, $\hat{\lambda}_\rho$ falls by more than half and becomes statistically insignificant in LC and MC.

The point estimates increase monotonically with the correlation between θ_a and θ_b , from $\hat{\lambda}_{0.2} = 0.169$ to $\hat{\lambda}_{0.5} = 0.262$ and $\hat{\lambda}_{0.8} = 0.320$, supporting the assumption that naivety increases with ρ . The difference between the HC and LC estimates is significant in the full sample (p -value = 0.025). This monotonicity holds in both subsamples, where the gaps between estimates are of similar magnitude but no longer significant, as the smaller samples yield larger standard errors.²⁹

As an external validation, we apply the same estimator to the feedback, random-role treatment of Jin, Luca and Martin (2021), an environment very similar to our experiments but without a public signal. Setting the hypothetical public signal equal to the prior mean ($\theta_b = 3$), the implied naivety parameter is $\hat{\lambda}_0 \approx 0.164$, slightly below $\hat{\lambda}_{0.2}$ and consistent with the monotone pattern documented above.³⁰

²⁹In Table A8, we relax the restriction that the weights on the sophisticated and naive guesses sum to one. Two findings support the baseline specification: (i) $\hat{\lambda}_\rho^N \approx \hat{\lambda}_\rho$ in all treatments, and (ii) $\hat{\lambda}_\rho^N + \hat{\lambda}_\rho^S \approx 1$ in all treatments.

³⁰In the feedback random-role treatment of Jin, Luca and Martin (2021), the average private state following non-disclosure is 1.680 and the average receiver guess following non-disclosure is 1.897. Treating the prior mean as the

Table 7: Model Fit by Treatment and Public Signal

Treatment	θ_b	Actual Guess	Estimated Model		Prediction of Alternative Models		
			Prediction	Distance	$\lambda = 0$	$\lambda = 1$	Cursed Equil.
LC	1	1.389	1.399	0.010	1.480	1.000	1.703
LC	3	2.277	2.081	0.196	1.894	3.000	2.114
LC	5	2.407	2.466	0.059	1.951	5.000	2.239
MC	1	1.235	1.138	0.097	1.187	1.000	1.434
MC	3	2.406	2.427	0.021	2.224	3.000	2.460
MC	5	3.023	2.999	0.024	2.294	5.000	2.813
HC	1	1.147	1.074	0.073	1.109	1.000	1.214
HC	3	2.698	2.824	0.126	2.741	3.000	2.834
HC	5	3.386	3.339	0.047	2.557	5.000	3.295
Average distance (unweighted)				0.073	0.311	0.953	0.156
Average distance (weighted)				0.076	0.208	0.700	0.144

* Notes: Each row reports actual and predicted mean receiver guesses conditional on non-disclosure, by treatment and the value of the public signal θ_b . Column 4 reports the predicted receiver guess using the estimated parameters from our model ($\hat{\lambda}_{LC} = 0.169$, $\hat{\lambda}_{MC} = 0.262$, $\hat{\lambda}_{HC} = 0.320$), and Column 5 reports the distance between the predicted and actual guesses. The last three columns report predictions from alternative models: the fully sophisticated benchmark ($\lambda = 0$), the fully naive benchmark ($\lambda = 1$), and the cursed equilibrium of [Eyster and Rabin \(2005\)](#) with estimated parameters $\hat{\chi}_{LC} = 0.199$, $\hat{\chi}_{MC} = 0.304$, and $\hat{\chi}_{HC} = 0.361$.

The magnitudes of λ_ρ reveal the degree of receiver naivety. Importantly, even in *HC*, where $\rho = 0.8$, $\hat{\lambda}_{0.8} = 0.32$ falls well below the correlation parameter, so receivers continue to draw meaningfully on the sophisticated component of their guess. Receivers are therefore partially but not fully naive, and their guess upon non-disclosure still reflects inference about sender behavior rather than direct use of the public signal.

Table 7 reports the fit of the estimated model for each treatment-by-public-signal cell. The predicted guess matches the actual guess closely in most cells, with an average absolute deviation of 0.07 between the two. The model fit is tightest at $\theta_b = 5$, where the estimated λ_ρ successfully reproduces the rise in receiver guesses from 2.41 in *LC* to 3.39 in *HC*, with deviations of at most 0.06 across the three treatments. The largest deviation arises at $\theta_b = 3$ under *LC*, where the model under-predicts the actual guess by 0.20. Our model reduces the gap to actual guesses by roughly 75 percent relative to the sophisticated benchmark and by over 90 percent relative to the naive benchmark.

A natural alternative to our proposed model is the cursed equilibrium ([Eyster and Rabin, 2005](#)), in which receivers partially neglect the strategic content of sender behavior. A cursed receiver's guess upon non-disclosure is a convex combination of a sophisticated guess and the conditional expectation of the private state given the public signal, $g^{\text{cursed}}(\theta_b) = (1 - \chi_\rho) \mu_\rho(\theta_b) + \chi_\rho \mathbb{E}[\theta_a \mid \theta_b]$,

public-signal reference point yields $\hat{\lambda}_0 = (1.897 - 1.680)/(3 - 1.680) \approx 0.164$.

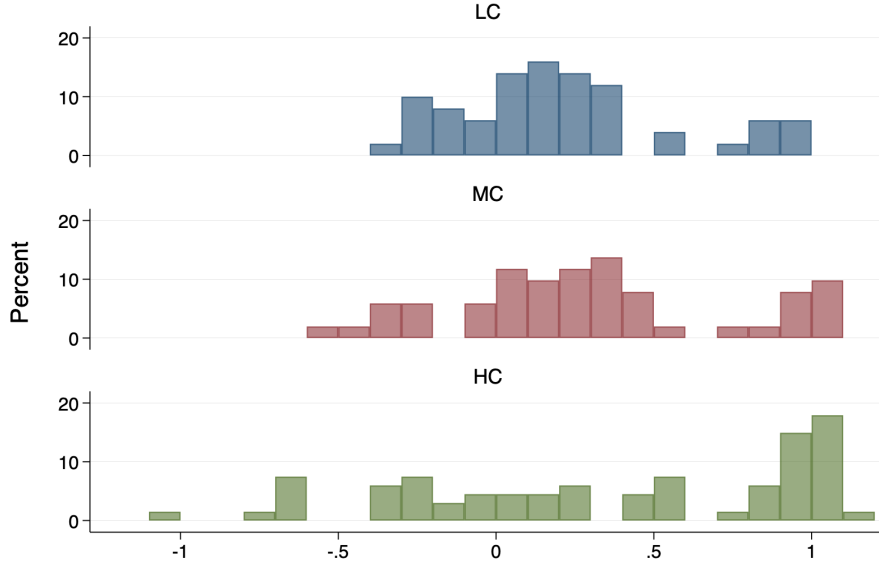


Figure 5: Distribution of Individual-Level Naivety

where $\mathbb{E}[\theta_a \mid \theta_b] = \rho \theta_b + (1 - \rho) \cdot 3$, and $\chi_\rho \in [0, 1]$ measures the degree of cursedness. Like our model, the cursed model nests full sophistication ($\chi_\rho = 0$), and a single parameter governs the departure from rationality.³¹ The two models differ in the prediction they make about the naive receiver's guess: a fully cursed receiver guesses $\mathbb{E}[\theta_a \mid \theta_b]$, whereas a fully naive receiver in our model guesses θ_b itself.

Estimating the cursed model yields $\hat{\chi}_{0.2} = 0.199$, $\hat{\chi}_{0.5} = 0.304$, and $\hat{\chi}_{0.8} = 0.361$. The predicted guesses are reported in the last column of Table 7. The cursed model out-performs the two corner benchmarks but does not match the fit of our model: the average distance to the actual guess is roughly 0.14–0.16, about twice the corresponding distance under our model, and our model fits more closely in seven of the nine treatment-signal cells. Forming $\mathbb{E}[\theta_a \mid \theta_b]$ requires the receiver to compute a conditional expectation from the joint distribution, while reading off θ_b requires no computation at all. Our specification therefore captures a less sophisticated and more cognitively plausible form of naivety, and the better fit suggests that receivers default to the simplest available signal.

Result 4. *Receivers overweight the public signal when interpreting non-disclosure, with the weight on the public signal $\hat{\chi}_\rho$ increasing monotonically in the correlation between the private state and the public signal. The fitted model predicts receiver guesses more accurately than the cursed equilibrium and the two corner benchmarks of full sophistication and full naivety.*

The treatment-level estimates of naivety summarize population averages but do not reveal

³¹One potential appeal of the cursed equilibrium is that the cursedness parameter is intended to be a primitive of the receiver and need not vary with ρ , so a single estimate would suffice across treatments. Here, we allow the cursedness parameter to vary with ρ for comparison purposes, since this would give us a better model fit under the cursed model.

heterogeneity across subjects. Figure 5 plots the distribution of subject-level estimates separately for each treatment, obtained by applying the estimator to each receiver individually.³² The distributions shift rightward as ρ rises, consistent with the treatment-level pattern. The modal $\hat{\lambda}$ is near 0.2 in *LC*, near 0.4 in *MC*, and near one in *HC*.

To quantify this heterogeneity, we partition receivers into four types by their estimated naivety level $\hat{\lambda}$: *counter-model* ($\hat{\lambda} < 0$), *sophisticated* ($0 \leq \hat{\lambda} < 0.2$), *moderately naive* ($0.2 \leq \hat{\lambda} < 0.8$), and *highly naive* ($\hat{\lambda} \geq 0.8$).³³ We do not restrict the individual-level naivety parameter to lie between zero and one. A negative $\hat{\lambda}$ arises when the receiver’s guess responds less to the public signal than the rational benchmark does (e.g. when the guess decreases in public signal, hence “counter-model”). The composition of types shifts with the correlation, and the shift is asymmetric across treatments. From *LC* to *MC*, the distribution moves relatively smoothly to the right: the moderately naive share rises modestly from 32 to 37 percent and the highly naive share rises from 12 to 20 percent, while the sophisticated share contracts from 30 to 22 percent. From *MC* to *HC*, by contrast, the distribution polarizes sharply: the highly naive share jumps from 20 to 40 percent, the sophisticated share collapses from 22 to 9 percent, and the moderately naive share reduces to 19 percent. Receiver naivety therefore does not reflect uniform behavior across subjects but a mixture, with the share of highly naive receivers expanding sharply at the highest correlation between the private state and the public signal.

5.4 Information Transmission

Now, we discuss the implications of sender and receiver behavior for information transmission. The key question is how much information receivers extract from non-disclosure. We compare the information made available by sender strategies to the information actually extracted by receivers. While sender strategies make substantial information available in equilibrium, receivers may fail to fully utilize this information.

We measure inference accuracy as $\mathcal{Q} = -\mathbb{E}[|\hat{\theta}_a - \theta_a|^{1.4} \mid \theta_b, m = \emptyset]$, computed over the non-disclosure sample, and informativeness as $\mathcal{I} = -\mathbb{E}[|\hat{\theta}_a - \theta_a|^{1.4} \mid \theta_b]$, computed over the full sample. Both are affine transformations of the expected receiver payoff. Higher values indicate that the receiver learns more effectively about the private state. The two differ in what they hold fixed: $\mathcal{Q}$ isolates the quality of inference when the sender withholds the state, whereas $\mathcal{I}$ also reflects how often withholding occurs.

We compute $\mathcal{Q}$ and $\mathcal{I}$ under three specifications, which together separate sender-side and receiver-side contributions to information transmission. All three specifications use the empirical sender strategy and differ only in how the receiver forms beliefs upon non-disclosure. The first specification, *Fully Naive*, assumes a fully naive receiver who always guesses $\hat{\theta}_a = \theta_b$. This

³²We restrict the sample to subjects with sufficient variation in the public signal upon non-disclosure as receivers. Specifically, a subject is kept if she has at least one non-disclosure observation at each of $\theta_b \in \{1, 3, 5\}$ and at least six non-disclosure observations as a receiver in total (out of 50 rounds). This leaves us with 70.6 percent of all subjects in the analysis sample.

³³Table A9 reports the share of subjects of each type in each treatment.

is equivalent to assuming $\lambda_\rho = 1$. The second specification, *Information Transmitted*, assumes a Bayesian receiver who correctly accounts for the selection in disclosure; it measures the information that the sender’s behavior makes available to an idealized receiver. This is equivalent to assuming $\lambda_\rho = 0$. The third specification, *Information Processed*, uses the actual receiver guesses in our data and captures how much of that information is extracted in practice. This reflects the set of λ_ρ (for each level of ρ) estimated from our experimental data. The gap between *Information Transmitted* and *Information Processed* measures the extent to which receivers fail to extract the information that the sender’s strategy makes available.

In Figure 6, we present inference accuracy and informativeness measured from all 50 rounds of our experiment. Figure A.4 shows the results when we restrict to the final 20 rounds. Despite the decline in receiver naivety observed in Table 6, the patterns are largely similar in these two figures.

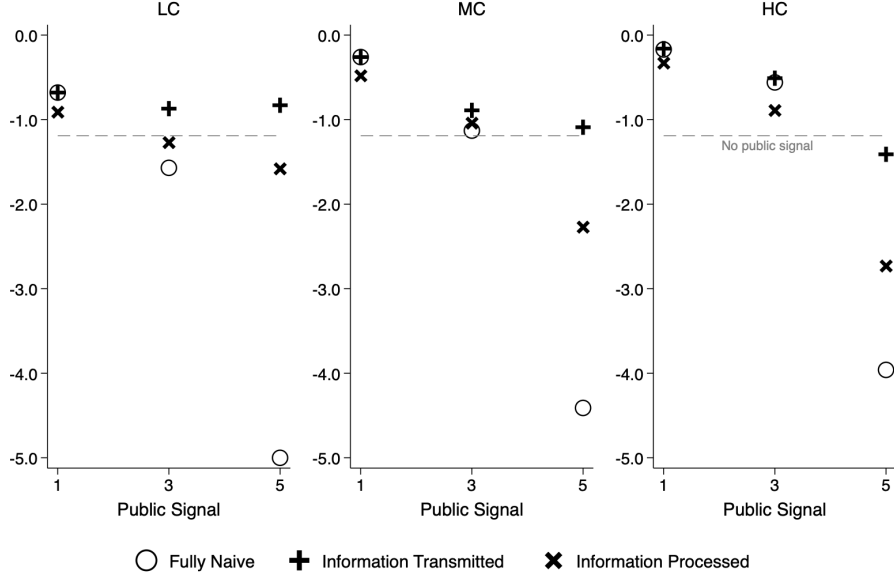
5.4.1 Inference Accuracy

Panel (a) of Figure 6 reports inference accuracy conditional on non-disclosure, by treatment and the public signal. Under *Fully Naive*, inference accuracy declines monotonically in θ_b and increases monotonically in ρ , conditional on θ_b . The decline in θ_b reflects selection into non-disclosure: senders with low θ_a are disproportionately likely to withhold, so the naive guess $\hat{\theta}_a = \theta_b$ becomes increasingly inaccurate as the public signal rises. The increase in ρ reflects the fact that higher correlation concentrates θ_a around θ_b , making the naive guess more accurate.

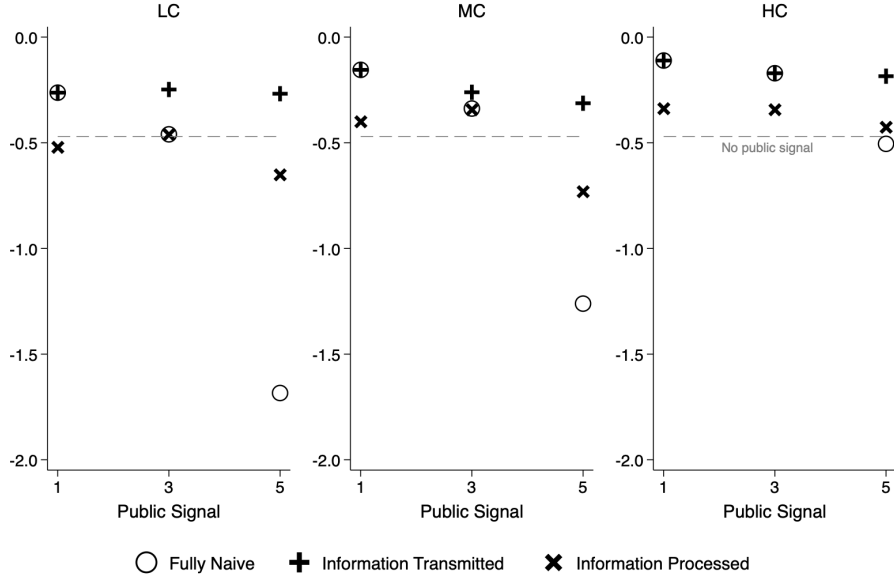
Information Transmitted is roughly flat in θ_b under LC, but declines in θ_b under MC and HC. Thus, when the public signal is high, inference from non-disclosure becomes less accurate in the higher-correlation treatments. Across treatments, its dependence on ρ is non-monotone. At $\theta_b = 1$, *Information Transmitted* is higher under MC and HC than under LC: because the public signal is already low, higher correlation makes low private states more likely among non-disclosures, so the act of withholding is easier to interpret. A similar, though less monotone, pattern appears at $\theta_b = 3$. By contrast, when $\theta_b = 5$, *Information Transmitted* falls as correlation rises. In this case, higher correlation makes relatively high and intermediate private states more likely to appear in the non-disclosure pool. Non-disclosure therefore becomes a noisier signal of a low private state, even though θ_a is more tightly linked to θ_b overall.

Information Processed declines as θ_b rises within each treatment. The decline is especially steep in the higher-correlation treatments, indicating that receivers make particularly poor use of non-disclosure when the public signal is high and the private and public signals are strongly correlated. Across treatments, the effect of correlation depends on the public signal. At $\theta_b = 1$ and $\theta_b = 3$, stronger correlation improves *Information Processed*: receivers extract more information from non-disclosure under MC and HC than under LC. At $\theta_b = 5$, stronger correlation sharply reduces *Information Processed*, and the lowest value occurs under HC.

The comparison between *Information Transmitted* and *Information Processed* provides evidence of receiver-side inefficiency. In every treatment-signal cell, *Information Processed* lies below *Infor-*



(a) Inference Accuracy (Non-Disclosure Sample)



(b) Informativeness (Full Sample)

Figure 6: Inference Accuracy and Informativeness by Treatment and Public Signal

Note: This figure presents the inference accuracy, $\mathcal{Q} = -\mathbb{E}[|\hat{\theta}_a - \theta_a|^{1.4} \mid \theta_b, m = \emptyset]$, and informativeness $\mathcal{I} = -\mathbb{E}[|\hat{\theta}_a - \theta_a|^{1.4} \mid \theta_b]$ under the indicated specification. Fully naive: sender behaves according to the empirical sender disclosure decisions, and the receiver always guesses $\hat{\theta}_a = \theta_b$. Information transmitted: sender behaves according to the empirical sender disclosure decisions, and the receiver makes a guess that minimizes expected loss. The optimal guesses are (1.0, 2.0, 2.0) in LC, (1.0, 2.0, 2.0) in MC, and (1.0, 3.0, 2.5) in HC, for $\theta_b = 1, 3, 5$ respectively. Information processed: the sample average of $-|\hat{\theta}_a - \theta_a|^{1.4}$.

mation Transmitted. Thus, receivers do not fully extract the information made available by the sender’s disclosure decision. The gap is small when $\theta_b = 1$, but becomes much larger when $\theta_b = 5$, reaching its maximum under *HC*. This pattern shows that the main failure is not that non-disclosure contains no information; rather, receivers fail to use that information when the public signal is high.

Figure 6 also depicts the baseline case of no public signal. More specifically, we compute *Information Processed* using the feedback and random-role treatment of Jin, Luca and Martin (2021), which provides no public signal and yields $\mathcal{Q} \approx -1.19$. At $\theta_b = 1$, *Information Processed* exceeds the no-public-signal benchmark in all three treatments (-0.91 , -0.48 , and -0.33 in *LC*, *MC*, and *HC*, respectively), so a low public signal helps receivers learn the private state. At $\theta_b = 3$, *LC* performs similarly to the no-public-signal environment (-1.27), while *MC* and *HC* achieve modest improvements (-1.04 and -0.89). At $\theta_b = 5$, by contrast, *Information Processed* falls well below the benchmark in all three treatments (-1.58 , -2.27 , and -2.73), and the loss is largest under *HC*, where receiver naivety is most pronounced. The public signal therefore helps receivers when it points to the lowest possible private state but backfires when it is favorable, and the magnitude of the backfiring rises with the correlation between the private state and the public signal.

In comparison, information transmitted by senders is above the no-public-signal benchmark except when the correlation is *HC* and the public signal is very favorable ($\theta_b = 5$). In that situation, senders’ strong strategic incentive to withhold the private state dominates the information conveyed by the public signal, and thus the information transmitted by senders is worse than what they would have transmitted without any public signal. In other words, backfire occurs in both sender behavior and receiver beliefs if the public signal is sufficiently favorable and the correlation between public and private information is sufficiently high.

5.4.2 Informativeness

Turning to the full-sample informativeness results in Panel (b) of Figure 6, several of the patterns from inference accuracy remain visible, but the magnitudes are substantially compressed. This compression follows mechanically from the fact that the full-sample measure averages over both disclosure and non-disclosure rounds. Since disclosure rounds generate zero loss, the full-sample measure is essentially the non-disclosure measure weighted by the probability of non-disclosure.

This distinction is most important when θ_b is high and correlation is strong. In the non-disclosure sample, the largest receiver-side losses occur at $\theta_b = 5$ under *MC* and especially under *HC*. In the full sample, these losses are much smaller because disclosure is very frequent in exactly these cells. Thus, although receivers make large errors conditional on observing non-disclosure, those errors receive relatively little weight in the full-sample average. As a result, the sharp deterioration in *Information Processed* at $\theta_b = 5$ under *HC* is muted in Panel (b).

The comparison to the no-public-signal benchmark, $\mathcal{I} \approx -0.47$, changes for the same reason. When the public signal and correlation are low or moderate, *Information Processed* is close to the no-public-signal benchmark and often above it, so public information weakly improves aggregate

information processing. When θ_b is high, the non-disclosure sample exhibits large information losses. In the full sample, *Information Processed* remains below the no-public-signal benchmark under *LC* and *MC*, so the backfire in overall informativeness is concentrated at low and medium correlation and is largest under *MC*. Under *HC*, by contrast, disclosure is so frequent that the losses conditional on non-disclosure receive little weight, and *Information Processed* lies slightly above the benchmark (at -0.43). The full-environment loss is therefore non-monotone in the correlation: it increases from *LC* to *MC* and vanishes at *HC*.

In comparison, the relative patterns change little from the non-disclosure sample to the full sample when the public signal is medium ($\theta_b = 3$). As shown in Figure 6(a), *Information Processed* can depart from *Information Transmitted* conditional on non-disclosure. At the same time, a stronger correlation may encourage the sender to withhold more when the private state is well below the public signal, but this effect on the disclosure probability is modest because a medium public signal conveys neither favorable nor unfavorable news by itself.

Overall, the full-sample informativeness figure reinforces the interpretation from the non-disclosure sample inference accuracy while clarifying the role of disclosure. Receiver-side inefficiency remains visible: *Information Processed* continues to lie below *Information Transmitted* in every cell. However, disclosure substantially attenuates the aggregate consequences of this inefficiency, especially when the public signal is high and the correlation is strong. These laboratory findings are consistent with the general patterns of informativeness predicted by our theoretical framework (Section 3).

Result 5. *Following non-disclosure, receivers extract less information than is implied by the sender’s strategy. This gap widens with the correlation between the private state and the public signal, particularly when the public signal is favorable. When correlation is strong and the public signal is favorable, receiver performance can fall well below the benchmark of voluntary disclosure without any public signal. In the full environment, however, high disclosure rates attenuate the aggregate information loss from receiver naivety.*

6 Implications and Conclusion

This paper has studied how voluntary disclosure responds to a correlated public signal when receivers are partially naive, and how the introduction of such a signal can reduce rather than enhance the total information available in a market. We developed a tractable disclosure framework that nests classical unraveling and the no-public-signal benchmark of [Jin, Luca and Martin \(2021\)](#) as limiting cases. It yields a closed-form decomposition of the equilibrium disclosure threshold into a baseline failure of unraveling independent of the public signal, and a signal-contingent component proportional to the gap between the signal and the prior mean. We then took the model to the lab in a 50-round disclosure experiment with 238 subjects across three correlation treatments. The data confirm the model’s qualitative predictions: voluntary disclosure declines in the public signal, the decline is amplified by the correlation between the private state and the public signal, and receiver guesses upon non-disclosure depart from the rational benchmark in the direction

of the public signal. The estimated naivety parameter rises monotonically with the correlation, and total informativeness can fall below the no-public-signal benchmark when the public signal is favorable, with the loss largest at intermediate correlation.

These results carry policy and business implications that extend well beyond the lab. We highlight three.

Disclosure policy depends on public signals. The classical debate over whether to mandate and standardize the disclosure of private quality information tends to focus on the private information itself, weighing policy intervention against market-driven voluntary disclosure of that information. Our study highlights how voluntary disclosure incentives depend on correlated public signals already available in the market. At first glance, one might argue that the higher the correlation between the public signal and the private information, the less concerning is strategic concealment, because consumers can learn much of the private information from the correlated public signal directly. We show that this intuition does not always hold: the role of correlation in total information is non-linear, because a favorable public signal encourages strategic concealment, and the tension between strategic concealment and the information directly observable in the public signal peaks when the correlation is meaningful but not exceedingly high. To the extent that policymakers care about the total amount of information available in the market—both the public signal and the selective disclosure of the private information—disclosure policy on the private information must account for prevailing public signals and their correlation with the focal private information.

The ongoing debate over whether colleges should adopt a test-required, test-optional, or test-blind policy for applicants' standardized test scores is a prime example. Part of the debate centers on the correlation between GPA (a public signal) and SAT/ACT scores (private information), but it largely ignores the strategic incentives to disclose or hide SAT/ACT scores at test-optional colleges. Proponents of test-optional or test-blind policies argue that SAT/ACT scores add little *new* information beyond GPA and other public signals (school district, demographics, family background, etc.) already observable to admissions offices, either because the scores reflect noise or because they are highly correlated with those public signals (Rothstein, 2004; Geiser, 2020). Proponents of test-required policies show that SAT/ACT scores improve predictions of college performance and post-college labor-market outcomes even after controlling for high-school GPA and other student observables (Bettinger, Evans and Pope, 2013; Sanchez and Comeaux, 2020; Chetty, Deming and Friedman, 2023; Cascio et al., 2024).³⁴ Both sides focus on statistical correlation, which is about 0.5 between high-school GPA and SAT/ACT scores in the raw data. Interestingly, we show that a mid-range correlation between the public signal and the private state (0.5, versus 0.2 and 0.8 in our experiment) is exactly the dangerous zone in which strategic silence behind a favorable public signal can crowd out the public signal so severely that the total information par-

³⁴On the theory side, Dessein, Frankel and Kartik (2025b) show that a college that can commit to how it uses information can never be made worse off by observing scores, so that any rationale for test-optional admissions must rest on outside pressures or a failure of standard rationality or commitment assumptions; Dessein, Frankel and Kartik (2025a) supply one such rationale, modeling test-optional policies as a response to social pressure on admission decisions.

tially naive consumers can process is worse with the public signal than without it. This finding is a cautionary message: disclosure policies must look beyond the focal private information and account for correlated public signals.³⁵

Consumer education is key. Because information crowding out and crowding in arise only when receivers are naive about the true meaning of non-disclosure, a second immediate policy implication is to educate consumers, investors, patients, and college admissions officers about the strategic incentives behind silence. Our results suggest that such education would be more effective still if it emphasized the role public signals play in shaping those incentives.

Unfortunately, consumer education is easier said than done. [Jin, Luca and Martin \(2021\)](#) find that receivers depart further from the equilibrium behavior predicted by classical unraveling theory than senders do, even when the two roles are played by the same subjects—a pattern our experiment confirms. They attribute this asymmetry to the greater strategic complexity of the receiver’s problem: inferring quality from silence requires hypothetical, conditional reasoning about the sender’s disclosure strategy rather than a direct best response. Consequently, receiver naivety erodes only slowly and only under the most favorable learning conditions, namely random role switching combined with immediate, direct, round-by-round feedback on the true state and the sender’s choice. Convergence slows sharply when feedback is absent or provided only in aggregate form, and the decline in receiver mistakes loses statistical significance altogether when roles are fixed. If receivers struggle to learn the meaning of silence even in this stark setting, educating consumers about how that meaning depends on correlated public signals is likely more challenging. The partial learning that receivers demonstrate in late rounds of our experiment is encouraging, but a substantial gap between receivers’ actual guesses and their implied beliefs persists even after 30 rounds. How to meet this challenge in the field deserves further study.

Correlated public signals may backfire. Our study does not speak to why the public signal exists in the first place. In reality, public signals are pervasive and endogenous. They may arise from mandatory disclosure laws (e.g., SEC financial disclosure requirements for publicly listed firms), firms’ voluntary disclosure in earlier periods (e.g., a CEO’s earnings forecast alongside mandatory SEC filings), or third-party reports (e.g., academic research or industry association reports). Whatever the source, we show that public signals shape the incentives to disclose or conceal private information.

These downstream effects can generate unintended consequences when regulators mandate disclosure of certain information and thereby define it into a public signal. Whether and how a listed firm strategically provides earnings guidance, for instance, depends on which accounting metrics SEC filings require and how favorable the firm looks on them. Although each mandate expands the supply of correlated public signals, our framework and experiment caution that this need not make the environment more informative overall. A signal related to, yet distinct from, the underlying object of interest can sharpen firms’ incentives to withhold private information

³⁵Relatedly, [Dessein, Frankel and Kartik \(2025a\)](#) model how colleges impute missing scores, including the non-adverse-inference rule that many test-optional colleges announce; such announcements institutionalize precisely the generous treatment of silence that arises endogenously from receiver naivety in our setting.

precisely when the public signal is favorable, leaving receivers with less—or more complex—total information than they would have had without the mandate.

The lesson is not that such mandates are always counterproductive; when the public signal is genuinely informative and receivers are sufficiently sophisticated, the direct informational gain can dominate. It is that a signal appearing to expand transparency can instead become a shield for selective non-disclosure, with the net effect depending on the correlation between the private and public states, the realized value of the public signal, and the degree of receiver naivety in the relevant population.

The same caution applies to private third parties that release correlated public signals such as academic researchers, news media, think tanks, and industry associations. A published rating, accreditation score, or sustainability index can shift the equilibrium of voluntary disclosure on adjacent dimensions in ways the releasing body does not anticipate. The concern deepens if senders can influence the public signal itself, through lobbying, choice of certifier, framing, metric definition, or strategic timing: they then have an incentive to engineer signals that are both favorable and apparently highly correlated with the private state. Both features broaden the set of intermediate states a sender can profitably conceal, and both operate through receiver naivety rather than any rational updating channel. These private incentives may in turn interact with the mandatory reporting policy, producing a complex strategic game among regulators, the firms they regulate, and the third parties that collect and disseminate related signals.

Limitations and future work. Several limitations point to directions for future research.

First, we model receiver naivety as anchoring on the public signal itself. Although this simple structure fits our data better than full unraveling, full naivety, and cursed equilibrium, richer behavioral specifications remain to be explored—including the cross-subject heterogeneity we document, endogenous learning over time, and asymmetric responses to favorable and unfavorable signals. A natural extension is to further distinguish our model from alternative behavioral models in explaining both sender and receiver behavior.

Second, our design fixes the joint distribution of the private and public states and assumes payoffs depend solely on the perceived and true values of the private state. Relaxing these restrictions—by embedding the disclosure game in a market with prices, repeated interaction, or competing senders, or by letting the public signal enter the receiver’s payoff function directly—would help connect our results to field settings such as credit ratings, earnings forecasts, environmental disclosures, hospital quality reporting, and college admissions.

Third, we document the existence and magnitude of receiver naivety rather than correcting it, leaving open the design of interventions that could reduce naivety in a world with both public signals and discretionary private information.

Finally, we treat the public signal as exogenous, whereas in many of these markets it is generated by the government, a private intermediary, or the sender herself, whose incentives shape the joint distribution of public and private information. Endogenizing the public signal is an important next step, particularly because the credibility of ratings, inspections, and accreditations is

itself a function of how much information they crowd out from voluntary disclosure.

References

- Acharya, Viral V., Peter DeMarzo, and Ilan Kremer.** 2011. "Endogenous Information Flows and the Clustering of Announcements." *American Economic Review*, 101(7): 2955–2979.
- Aghamolla, Cyrus, and Byeong-Je An.** 2021. "Voluntary Disclosure with Evolving News." *Journal of Financial Economics*, 140(1): 21–53.
- Ahn, Shinhye, Theresa S. Cho, and Cecile K. Cho.** 2025. "Stock Market Reaction to Environmental Performance: The Role of Organizational Reputation as an Interpretive Lens." *Organization & Environment*, 38(3).
- Benjamin, Daniel J.** 2019. "Errors in probabilistic reasoning and judgment biases." *Handbook of Behavioral Economics: Applications and Foundations* 1, 2: 69–186.
- Bettinger, Eric P., Brent J. Evans, and Devin G. Pope.** 2013. "Improving college performance and retention the easy way: Unpacking the ACT exam." *American Economic Journal: Economic Policy*, 5(2): 26–52.
- Beyer, Anne, Daniel A. Cohen, Thomas Z. Lys, and Beverly R. Walther.** 2010. "The Financial Reporting Environment: Review of the Recent Literature." *Journal of Accounting and Economics*, 50(2–3): 296–343.
- Board, Oliver.** 2009. "Competition and disclosure." *The Journal of Industrial Economics*, 57(1): 197–213.
- Buchner, Axel, Abdulkadir Mohamed, and Brahim Saadouni.** 2017. "The Association Between Earnings Forecast in IPOs Prospectuses and Earnings Management: An Empirical Analysis." *Journal of International Financial Markets, Institutions and Money*, 51: 92–105.
- Cascio, Elizabeth, Bruce Sacerdote, Doug Staiger, and Michele Tine.** 2024. "Report From Working Group on the Role of Standardized Test Scores in Undergraduate Admissions." *Dartmouth University*.
- Chen, Daniel L, Martin Schonger, and Chris Wickens.** 2016. "oTree—An open-source platform for laboratory, online, and field experiments." *Journal of Behavioral and Experimental Finance*, 9: 88–97.
- Chen, Marty Haoyuan, and Ginger Zhe Jin.** 2025. "Voluntary Report of Standardized Test Scores: An Experimental Study." National Bureau of Economic Research.
- Chetty, Raj, David J Deming, and John N Friedman.** 2023. "Diversifying society's leaders? The causal effects of admission to highly selective private colleges." National Bureau of Economic Research.
- Cheyne, Edwige.** 2013. "A Theory of Voluntary Disclosure and Cost of Capital." *Review of Accounting Studies*, 18(4): 987–1020.
- Cohen, Shani, and Shengwu Li.** 2026. "Sequential Cursed Equilibrium." *American Economic Review*, 116(3): 934–976.

- Dessein, Wouter, Alex Frankel, and Navin Kartik.** 2025a. "Test-optional admissions." *American Economic Review*, 115(9): 3130–3170.
- Dessein, Wouter, Alex Frankel, and Navin Kartik.** 2025b. "The test-optional puzzle." *AEA Papers and Proceedings*, 115: 669–675.
- Dranove, David, and Ginger Zhe Jin.** 2010. "Quality disclosure and certification: Theory and practice." *Journal of Economic Literature*, 48(4): 935–963.
- Dye, Ronald A.** 1985. "Disclosure of Nonproprietary Information." *Journal of Accounting Research*, 23(1): 123–145.
- Edwards, Ward.** 1968. "Conservatism in human information processing." *Formal Representation of Human Judgment*.
- Enke, Benjamin.** 2020. "What you see is all there is." *The Quarterly Journal of Economics*, 135(3): 1363–1398.
- Enke, Benjamin, and Florian Zimmermann.** 2019. "Correlation neglect in belief formation." *The review of economic studies*, 86(1): 313–332.
- Esponda, Ignacio, and Emanuel Vespa.** 2024. "Contingent Thinking and the Sure-Thing Principle: Revisiting Classic Anomalies in the Laboratory." *The Review of Economic Studies*, 91(5): 2806–2831.
- Eyster, Erik, and Matthew Rabin.** 2005. "Cursed equilibrium." *Econometrica*, 73(5): 1623–1672.
- Farina, Agata, and Clément Herman.** 2026. "Communicating with Data-Generating Processes: An Experimental Analysis." *Available at SSRN 6482578*.
- Farina, Agata, and Mario Leccese.** 2024. "Hiding a Flaw? Experimental Evidence on Multi-Dimensional Information Disclosure."
- Farina, Agata, Guillaume Fréchette, Alessandro Ispano, Alessandro Lizzeri, and Jacopo Perego.** 2024. "The selective disclosure of evidence: An experiment." National Bureau of Economic Research.
- Feltovich, Nick, Richmond Harbaugh, and Ted To.** 2002. "Too cool for school? Signalling and countersignalling." *RAND Journal of Economics*, 630–649.
- Fong, Meng-Jhang, Po-Hsuan Lin, and Thomas R Palfrey.** 2025. "Cursed sequential equilibrium." *American Economic Review*, 115(8): 2616–2658.
- Fréchette, Guillaume R, Alessandro Lizzeri, and Jacopo Perego.** 2022. "Rules and commitment in communication: An experimental analysis." *Econometrica*, 90(5): 2283–2318.
- Frenkel, Sivan, Ilan Guttman, and Ilan Kremer.** 2020. "The effect of exogenous information on voluntary disclosure and market quality." *Journal of Financial Economics*, 138(1): 176–192.
- Gabaix, Xavier, and David Laibson.** 2006. "Shrouded attributes, consumer myopia, and information suppression in competitive markets." *The Quarterly Journal of Economics*, 121(2): 505–540.
- Geiser, Saul.** 2020. "SAT/ACT Scores, High-School GPA, and the Problem of Omitted Variable Bias: Why the UC Taskforce's Findings Are Spurious. Research & Occasional Paper Series: CSHE. 1.2020." *Center for Studies in Higher Education*.
- Grossman, Sanford J.** 1981. "The informational role of warranties and private disclosure about

- product quality." *The Journal of Law and Economics*, 24(3): 461–483.
- Grossman, Sanford J., and Oliver D. Hart.** 1980. "Disclosure Laws and Takeover Bids." *The Journal of Finance*, 35(2): 323–334.
- Gurney, Nikolos, and George Loewenstein.** 2020. "Filling in the Blanks: What Restaurant Patrons Assume About Missing Sanitation Inspection Grades." *Journal of Public Policy & Marketing*, 39(3).
- Guttman, Ilan, Ilan Kremer, and Andrzej Skrzypacz.** 2014. "Not Only What but Also When: A Theory of Dynamic Voluntary Disclosure." *American Economic Review*, 104(8): 2400–2420.
- Hagenbach, Jeanne, and Eduardo Perez-Richet.** 2018. "Communication with evidence in the lab." *Games and Economic Behavior*, 112: 139–165.
- Heidhues, Paul, Botond Köszegi, and Takeshi Murooka.** 2016. "Inferior products and profitable deception." *The Review of Economic Studies*, 84(1): 323–356.
- Hirshleifer, David, and Siew Hong Teoh.** 2003. "Limited attention, information disclosure, and financial reporting." *Journal of Accounting and Economics*, 36(1-3): 337–386.
- Ito, Hiroto, and H. Sugawara.** 2005. "Relationship Between Accreditation Scores and the Public Disclosure of Accreditation Reports: A Cross Sectional Study." *Quality & Safety in Health Care*, 14(2): 87–92.
- Jin, Ginger Zhe, Michael Luca, and Daniel Martin.** 2021. "Is no news (perceived as) bad news? An experimental investigation of information disclosure." *American Economic Journal: Microeconomics*, 13(2): 141–173.
- Jin, Ginger Zhe, Michael Luca, and Daniel Martin.** 2022. "Complex disclosure." *Management Science*, 68(5): 3236–3261.
- Jovanovic, Boyan.** 1982. "Truthful Disclosure of Information." *The Bell Journal of Economics*, 13(1): 36–44.
- Jung, Woon-Oh, and Young K. Kwon.** 1988. "Disclosure When the Market Is Unsure of Information Endowment of Managers." *Journal of Accounting Research*, 26(1): 146–153.
- Kallapur, Sanjay, Abdul Khizer, Hariom Manchiraju, and Rajesh Vijayaraghavan.** 2025. "Does Observability of Ratings Shopping Improve Ratings Quality?" *Journal of Accounting and Economics*, 80(1).
- Kim, Jung Min, Daniel J. Taylor, and Robert E. Verrecchia.** 2021. "Voluntary Disclosure When Private Information and Disclosure Costs Are Jointly Determined." *Review of Accounting Studies*, 26(3): 971–1001.
- Mählmann, Thomas.** 2008. "Rating Agencies and the Role of Rating Publication Rights." *Journal of Banking & Finance*, 32(11): 2412–2422.
- Marquis, Christopher, Michael W. Toffel, and Yanhua Zhou.** 2016. "Scrutiny, Norms, and Selective Disclosure: A Global Study of Greenwashing." *Organization Science*, 27(2): 483–504.
- Martínez-Marquina, Alejandro, Muriel Niederle, and Emanuel Vespa.** 2019. "Failures in contingent reasoning: The role of uncertainty." *American Economic Review*, 109(10): 3437–3474.
- McCormick, Danny, David U. Himmelstein, Steffie Woolhandler, Sidney M. Wolfe, and David H. Bor.** 2002. "Relationship Between Low Quality-of-Care Scores and HMOs' Subsequent

- Public Disclosure of Quality-of-Care Scores." *JAMA*, 288(12): 1484–1490.
- McManus, Brian, Jessica Howell, and Michael Hurwitz.** 2023. "Strategic Disclosure of Test Scores: Evidence from US College Admissions. EdWorkingPaper No. 23-843." *Annenberg Institute for School Reform at Brown University*.
- Milgrom, Paul R.** 1981. "Good News and Bad News: Representation Theorems and Applications." *The Bell Journal of Economics*, 12(2): 380–391.
- Montero, Maria, and Jesal D Sheth.** 2021. "Naivety about hidden information: An experimental investigation." *Journal of Economic Behavior & Organization*, 192: 92–116.
- Morris, Stephen, and Hyun Song Shin.** 2002. "Social value of public information." *American Economic Review*, 92(5): 1521–1534.
- Mullainathan, Sendhil, Joshua Schwartzstein, and Andrei Shleifer.** 2008. "Coarse thinking and persuasion." *The Quarterly journal of economics*, 123(2): 577–619.
- Penczynski, Stefan, Christian Koch, and Sihong Zhang.** 2022. "Disclosure of verifiable information under competition: an experimental study."
- Quigley, Daniel, and Ansgar Walther.** 2020. "Inside and outside information: reverse unraveling and stress test design." *Unpublished Working Paper*.
- Rappoport, Daniel.** 2025. "Evidence and skepticism in verifiable disclosure games." *Theoretical Economics*, 20(4): 1213–1246.
- Rothstein, Jesse M.** 2004. "College performance predictions and the SAT." *Journal of Econometrics*, 121(1-2): 297–317.
- Sanchez, Henry, and Eddie Comeaux.** 2020. "Report of the UC Academic Council Standardized Testing Task Force." *University of California Systemwide Academic Senate*.
- Skinner, Douglas J.** 1994. "Why Firms Voluntarily Disclose Bad News." *Journal of Accounting Research*, 32(1): 38–60.
- Trautmann, Stefan T, and Gijs van de Kuilen.** 2015. "Belief elicitation: A horse race among truth serums." *The Economic Journal*, 125(589): 2116–2135.
- Verrecchia, Robert E.** 1983. "Discretionary Disclosure." *Journal of Accounting and Economics*, 5: 179–194.

SUPPLEMENTAL APPENDIX

A Figures and Tables

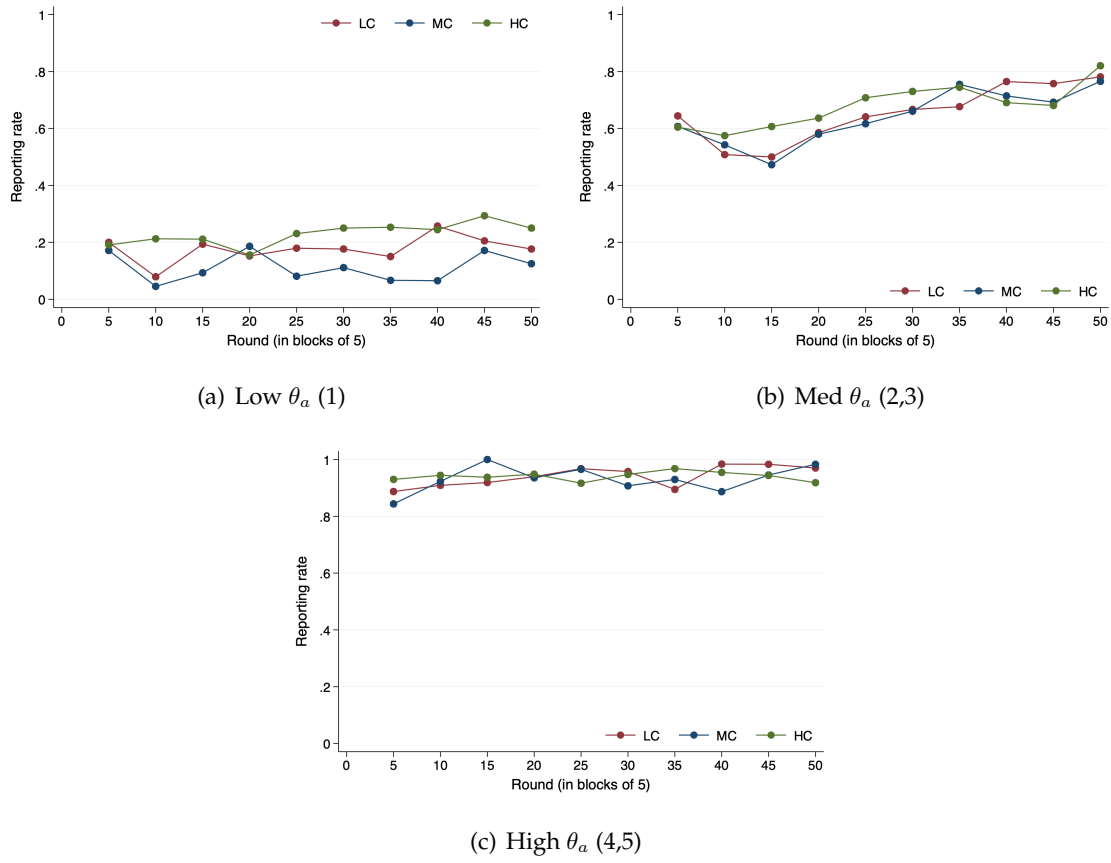


Figure A.1: Sender Disclosure Rates by Round

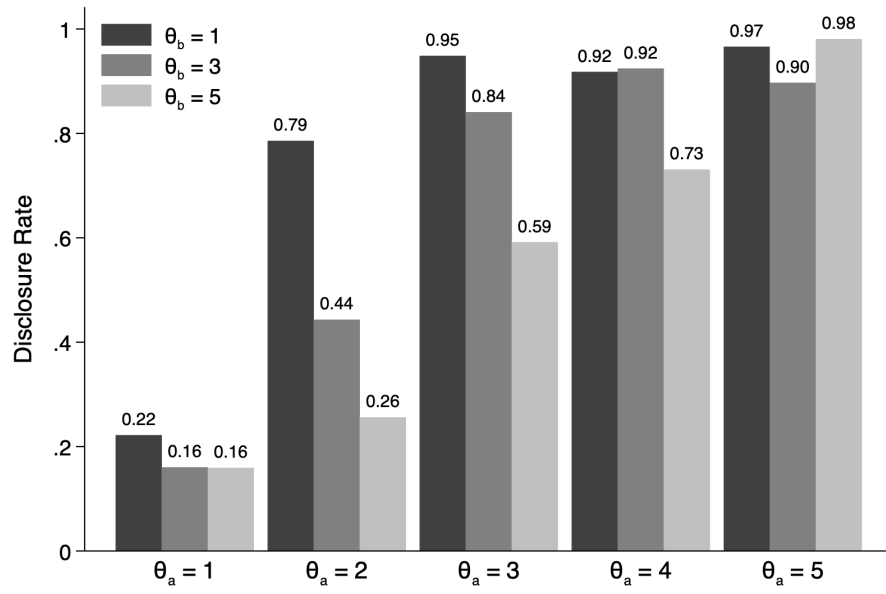


Figure A.2: Sender Disclosure Rates by Public Signal and Private State (Last 20 Rounds)

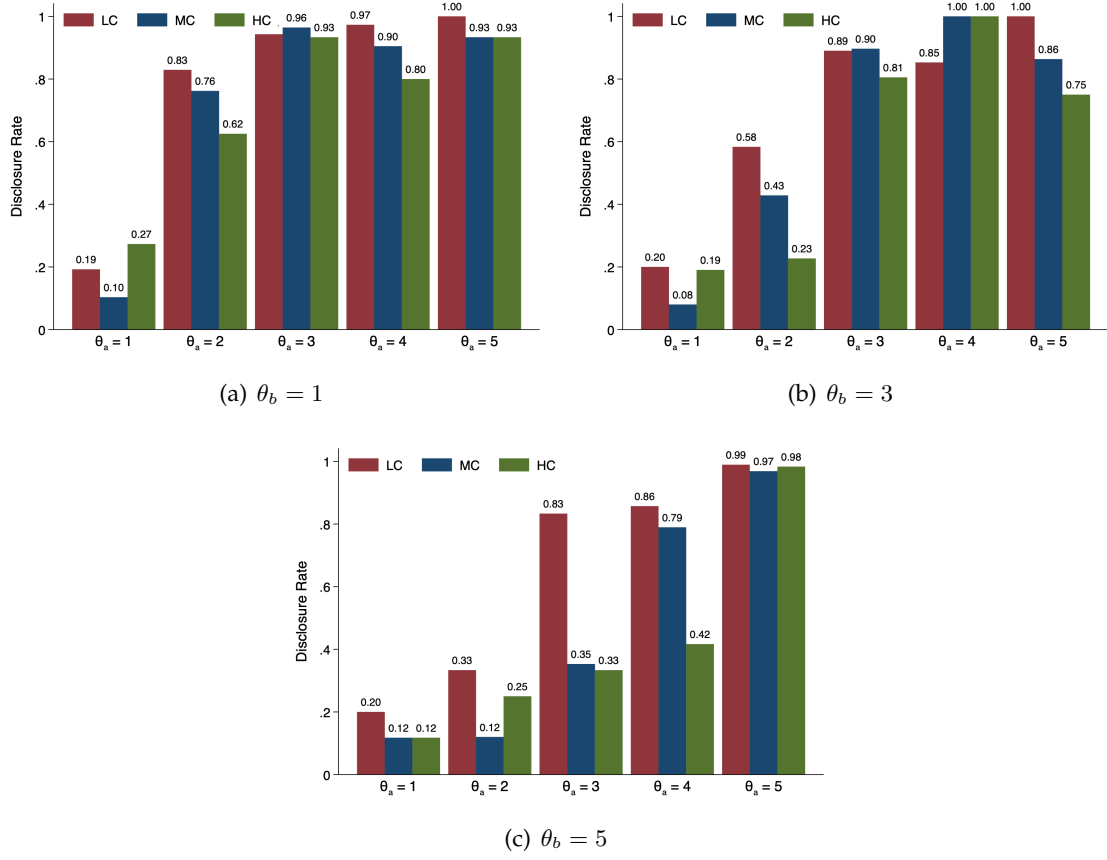
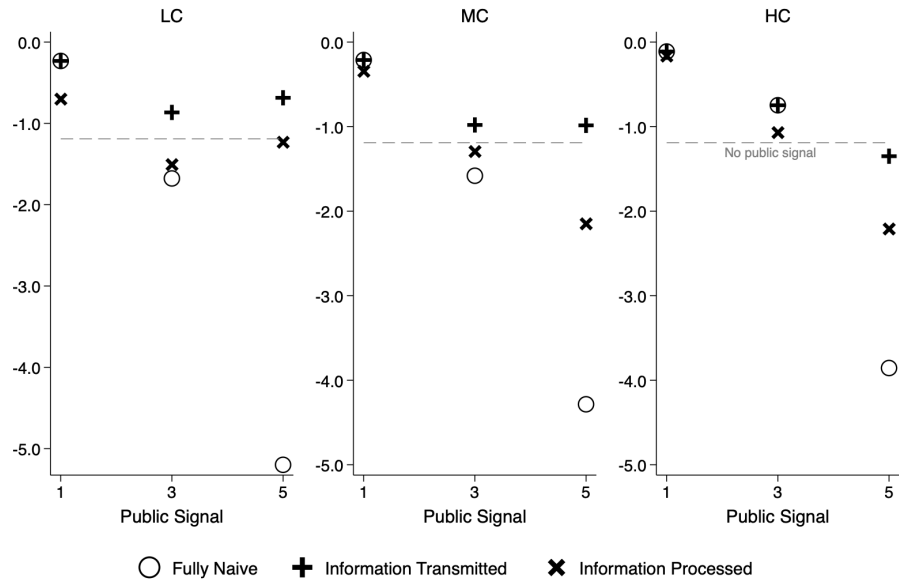
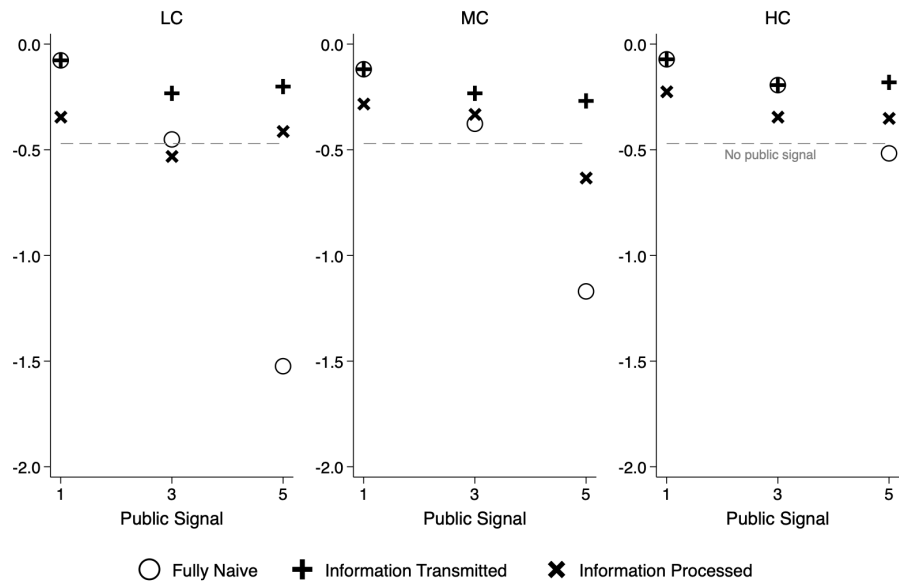


Figure A.3: Sender Disclosure Rates (Last 20 Rounds)



(a) Inference Accuracy (Non-Disclosure Sample)



(b) Informativeness (Full Sample)

Figure A.4: Inference Accuracy and Informativeness (Last 20 Rounds)

Note: This figure replicates Figure 6 while restricting the data to the last 20 rounds of the experiment.

Table A1: Subject Demographics

	LC	MC	HC
Age	20.7	19.9	20.2
Female	0.530	0.452	0.445
White	0.485	0.452	0.327
Black	0.106	0.113	0.155
Asian	0.348	0.435	0.455
Stem Major	0.803	0.774	0.773
Freshman	0.167	0.258	0.227
Sophomore	0.273	0.226	0.227
Junior	0.333	0.323	0.264
Senior	0.227	0.194	0.282
Friend in Session	0.091	0.097	0.082
Native Speaker	0.909	0.806	0.918
Risk Preference	4.288	4.548	4.182
Observations	66	62	110

Table A2: Pairwise Tests for Treatment Effects

	LC vs. MC	MC vs. HC	LC vs. HC
Panel A: Sender Disclosure Rates			
When $\theta_b = 1$			
$\theta_a = 1$	0.004	0.000	0.238
$\theta_a = 2$	0.782	0.490	0.332
$\theta_a = 3$	0.825	0.584	0.430
$\theta_a = 4$	0.671	0.246	0.066
$\theta_a = 5$	0.482	0.238	0.506
When $\theta_b = 3$			
$\theta_a = 1$	0.400	0.547	0.883
$\theta_a = 2$	0.003	0.447	0.001
$\theta_a = 3$	0.287	0.025	0.003
$\theta_a = 4$	0.320	0.128	0.440
$\theta_a = 5$	0.085	0.103	0.000
When $\theta_b = 5$			
$\theta_a = 1$	0.785	0.705	0.521
$\theta_a = 2$	0.073	0.710	0.262
$\theta_a = 3$	0.002	0.469	0.045
$\theta_a = 4$	0.121	0.164	0.004
$\theta_a = 5$	0.287	0.289	0.725
Panel B: Receiver Guesses Upon Non-Disclosure			
When $\theta_b = 1$	0.006	0.017	0.000
When $\theta_b = 3$	0.193	0.000	0.000
When $\theta_b = 5$	0.000	0.038	0.000

Table A3: Regressions on Learning and Stated Beliefs

	Sender disclosure (0/1)			Receiver guess (no report)		
	(1)	(2)	(3)	(4)	(5)	(6)
Round	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	-0.009** (0.004)	-0.010** (0.004)	-0.011* (0.005)
Round $\times$ MC	-0.001 (0.001)	-0.001 (0.001)	-0.001** (0.001)	-0.003 (0.004)	-0.003 (0.005)	0.003 (0.006)
Round $\times$ HC	-0.001* (0.001)	-0.001* (0.001)	-0.001** (0.001)	0.003 (0.004)	0.003 (0.004)	0.005 (0.006)
Belief of receiver guess $< \theta_a$	0.071* (0.035)	0.075** (0.035)				
Belief low $\times$ MC	0.103** (0.036)	0.104** (0.037)				
Belief low $\times$ HC	0.036 (0.040)	0.033 (0.040)				
Implied belief of θ_a (no report)				0.358*** (0.061)	0.340*** (0.062)	
Public signal (θ_b) FE	X	X	X	X	X	X
Treatment and private state (θ_a) FE	X	X		X	X	
Demographics		X			X	
Subject FE			X			X
Observations	5,950	5,950	5,950	2,158	2,158	2,177

Notes: Columns (1) to (3) report OLS estimates of the sender's disclosure decision, an indicator equal to one when the sender reports θ_a . Columns (4) to (6) report OLS estimates of the receiver's guess, restricted to rounds in which the sender did not report. *Belief of receiver guess $< \theta_a$* is an indicator equal to one when the sender's stated belief about the average receiver guess upon non-disclosure lies below her realized private state. *Implied belief of θ_a* is the posterior mean of θ_a conditional on non-disclosure, obtained by applying Bayes' rule to the receiver's stated beliefs about sender reporting rates. Standard errors, clustered at the session level, are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: Regressions on Changes in Receiver Guesses of Nonreported Numbers

Dependent variable: guess – last guess (no report)	$\theta_b = 1$		$\theta_b = 3$		$\theta_b = 5$	
	(1)	(2)	(3)	(4)	(5)	(6)
MC	-0.020 (0.054)		0.102 (0.365)		0.342 (0.318)	
HC	-0.046 (0.095)		0.305 (0.373)		0.831 (0.485)	
Round	0.003* (0.002)	0.000 (0.003)	0.007 (0.005)	0.011 (0.011)	0.007* (0.004)	0.010 (0.009)
Round x MC	-0.005** (0.002)	-0.001 (0.004)	0.001 (0.007)	-0.002 (0.013)	-0.002 (0.006)	-0.015 (0.017)
Round x HC	-0.003 (0.003)	-0.003 (0.005)	-0.009 (0.006)	-0.015 (0.011)	-0.007 (0.014)	-0.016 (0.026)
Implied belief of nonreported θ_a	0.022 (0.029)		0.016 (0.105)		0.178** (0.067)	
Overguessed	-0.516*** (0.025)	-0.718*** (0.108)	-0.112 (0.264)	-0.082 (0.774)	-0.658** (0.257)	-0.912** (0.355)
Overguessed x MC	0.147 (0.139)	-0.050 (0.191)	-0.315 (0.282)	-0.328 (0.798)	-0.043 (0.312)	-0.672 (0.475)
Overguessed x HC	-0.493*** (0.099)	-0.665** (0.233)	-0.176 (0.268)	-0.194 (0.793)	-0.696* (0.370)	0.222 (0.754)
Underguessed	-0.041 (0.037)	-0.099 (0.071)	0.375 (0.304)	0.315 (0.615)	0.197 (0.349)	0.207 (0.364)
Underguessed x MC	0.257*** (0.055)	0.239** (0.112)	-0.579* (0.313)	-0.483 (0.697)	-0.296 (0.414)	-0.347 (0.533)
Underguessed x HC	0.022 (0.054)	0.073 (0.092)	-0.224 (0.314)	-0.081 (0.659)	0.154 (0.544)	-0.309 (0.996)
Demographics	X		X		X	
Subject FE		X		X		X
Observations	896	896	367	368	255	266

Notes: The dependent variable is the change in a receiver's guess upon non-disclosure relative to her previous non-disclosure guess. *Overguessed* and *Underguessed* are indicators for whether the receiver's previous guess exceeded or fell short of the realized private state. *Implied belief of nonreported θ_a* is the posterior mean of θ_a conditional on non-disclosure, obtained by applying Bayes' rule to the receiver's stated beliefs about sender reporting rates. Standard errors, clustered at the session level, are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Summary of Receiver Guesses (Last 20 Rounds)

	LC		MC		HC	
	N	Guess	N	Guess	N	Guess
When $\theta_b = 1$						
Guess (report=1)	30	1.200	24	1.125	172	1.285
Guess (report=2)	68	2.176	32	2.000	10	2.000
Guess (report=3)	66	3.136	54	3.000	28	2.571
Guess (report=4)	72	4.042	38	3.605	24	3.750
Guess (report=5)	58	4.931	28	4.821	28	4.964
Guess (no report)	73	1.438	113	1.150	237	1.051
Guess $- \theta_a$ (no report)	73	0.247	113	0.000	237	-0.025
When $\theta_b = 3$						
Guess (report=1)	14	1.357	4	1.000	8	1.500
Guess (report=2)	42	2.286	18	2.167	10	2.200
Guess (report=3)	130	3.154	226	3.022	488	3.076
Guess (report=4)	58	4.103	40	4.075	24	3.958
Guess (report=5)	60	4.933	38	5.000	24	5.000
Guess (no report)	56	2.259	51	2.186	97	2.649
Guess $- \theta_a$ (no report)	56	0.438	51	0.206	97	0.093
When $\theta_b = 5$						
Guess (report=1)	14	1.357	4	2.000	4	1.000
Guess (report=2)	30	2.133	6	2.000	6	2.500
Guess (report=3)	60	3.050	12	3.000	12	3.250
Guess (report=4)	36	4.139	30	4.000	10	4.100
Guess (report=5)	188	4.963	246	4.959	588	4.966
Guess (no report)	68	2.059	56	2.679	48	3.271
Guess $- \theta_a$ (no report)	68	0.250	56	0.393	48	0.729

Table A6: Regressions on Sender Disclosures and Receiver Guesses (Last 20 Rounds)

	Sender disclosures (0/1)		Receiver guesses (no report)	
	(1)	(2)	(3)	(4)
MC	-0.061** (0.028)		0.145 (0.150)	
HC	-0.050 (0.029)		0.234 (0.145)	
θ_a	0.218*** (0.009)	0.213*** (0.006)		
θ_b	-0.043*** (0.006)	-0.039*** (0.006)	0.411*** (0.055)	0.412*** (0.064)
Demographics	X		X	
Subject FE		X		X
Observations	2,380	2,380	799	799

Notes: The sample is restricted to the final 20 rounds of the experiment. Columns (1) and (2) report OLS estimates of the sender's disclosure decision, an indicator equal to one when the sender reports θ_a . Columns (3) and (4) report OLS estimates of the receiver's guess, restricted to rounds in which the sender did not report. LC is the omitted treatment category. Standard errors, clustered at the session level, are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: Summary of Receiver Beliefs

	LC	MC	HC
When $\theta_b = 1$			
Reporting rate ($\theta_a = 1$)	0.147	0.160	0.273
Reporting rate ($\theta_a = 2$)	0.299	0.435	0.444
Reporting rate ($\theta_a = 3$)	0.584	0.665	0.625
Reporting rate ($\theta_a = 4$)	0.757	0.763	0.693
Reporting rate ($\theta_a = 5$)	0.854	0.817	0.756
When $\theta_b = 3$			
Reporting rate ($\theta_a = 1$)	0.107	0.070	0.127
Reporting rate ($\theta_a = 2$)	0.253	0.203	0.231
Reporting rate ($\theta_a = 3$)	0.660	0.684	0.723
Reporting rate ($\theta_a = 4$)	0.739	0.740	0.668
Reporting rate ($\theta_a = 5$)	0.806	0.776	0.693
When $\theta_b = 5$			
Reporting rate ($\theta_a = 1$)	0.105	0.068	0.100
Reporting rate ($\theta_a = 2$)	0.195	0.146	0.175
Reporting rate ($\theta_a = 3$)	0.473	0.410	0.333
Reporting rate ($\theta_a = 4$)	0.685	0.629	0.504
Reporting rate ($\theta_a = 5$)	0.934	0.929	0.926

Table A8: Estimation Results Without Restriction on Weights (All Rounds)

	LC	MC	HC
λ_{ρ}^S	0.891*** (0.094)	0.810*** (0.099)	0.655*** (0.079)
λ_{ρ}^N	0.145** (0.033)	0.227* (0.076)	0.333*** (0.068)
Observations	559	605	1,013

Notes: In these regressions, we relax the assumption that the weights on the sophisticated guess and the public signal sum to one, and estimate those weights. Standard errors are clustered at the session level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A9: Distribution of Receiver Types by Treatment

	Counter-model ($\hat{\lambda} < 0$)	Sophisticated ($0 \leq \hat{\lambda} < 0.2$)	Moderately naive ($0.2 \leq \hat{\lambda} < 0.8$)	Highly naive ($\hat{\lambda} \geq 0.8$)
LC ($N = 50$)	0.260	0.300	0.320	0.120
MC ($N = 51$)	0.216	0.216	0.373	0.196
HC ($N = 67$)	0.313	0.090	0.194	0.403

* *Notes:* The table reports the share of receivers in each treatment classified into one of four types based on the subject-level estimate of naivety. The sample is restricted to subjects with at least one non-disclosure observation at each of $\theta_b \in \{1, 3, 5\}$ and at least six non-disclosure observations as a receiver in total.

B Proofs

Proof of Proposition 1. In equilibrium, $\theta_a^*(\theta_b) = \hat{\theta}_a(\varnothing, \theta_b)$, so

$$\theta_a^*(\theta_b) = (1 - \lambda_\rho) \cdot \mathbb{E}[\theta_a | \theta_a \leq \theta_a^*(\theta_b), \theta_b] + \lambda_\rho \cdot \theta_b$$

By definition of a truncated mean, $\mathbb{E}[\theta_a | \theta_a \leq \theta_a^*(\theta_b), \theta_b] \leq \theta_a^*(\theta_b)$, with strict inequality whenever $\theta_a^*(\theta_b) > \theta_a^{min}$. Since $\theta_a^*(\theta_b)$ is a convex combination of this truncated mean and θ_b , it follows that $\theta_a^*(\theta_b) \leq \theta_b$, with strict inequality whenever $\theta_b > \theta_a^{min}$. Given that $\theta_a^*(\theta_b) < \theta_b$ and the conditional distribution of θ_a on $[\theta_a^{min}, \theta_a^*(\theta_b)]$ is uniform,

$$\mathbb{E}[\theta_a | \theta_a \leq \theta_a^*(\theta_b), \theta_b] = \frac{\theta_a^{min} + \theta_a^*(\theta_b)}{2}.$$

Substituting into the equilibrium condition and solving yields

$$\theta_a^*(\theta_b) = \frac{1 - \lambda_\rho}{1 + \lambda_\rho} \theta_a^{min} + \frac{2\lambda_\rho}{1 + \lambda_\rho} \theta_b$$

At $\theta_b = \theta_a^{min}$, the formula gives $\theta_a^*(\theta_a^{min}) = \theta_a^{min}$. ■

C Equilibrium of the Discrete Experimental Game

Section 3 derives the equilibrium on a continuum, whereas the experiment uses $\mathcal{A} = \{1, 2, 3, 4, 5\}$, $\mathcal{B} = \{1, 3, 5\}$, and a half-unit guess grid. This appendix verifies that Hypotheses 1 and 2 hold in the discrete game. As in the continuum, the non-disclosure pool never contains the public signal itself, so the equilibrium depends on (ρ, θ_b) only through (λ_ρ, θ_b) , and every equilibrium cutoff lies within $2/(1 + \lambda_\rho)$ of the threshold in Equation 3.

The section proceeds in four steps: we first set up the discrete game, then characterize its equilibrium set, next verify that Hypotheses 1 and 2 hold across every equilibrium selection, and finally show that the conclusions survive the restriction of guesses to the half-unit grid.

C.1 Setup

Fix $\rho \in (0, 1)$ and $\theta_b \in \mathcal{B}$, and write $\lambda \equiv \lambda_\rho \in [0, 1)$ and $q \equiv (1 - \rho)/5 > 0$, so that $\pi(\theta_a | \theta_b) = \rho \cdot \mathbb{1}[\theta_a = \theta_b] + q$ as in the design. Upon disclosure the receiver's unique optimal action is $\hat{\theta}_a = \theta_a$. Since U_S is strictly increasing in the guess, the sender discloses iff θ_a exceeds the non-disclosure guess $\hat{\theta}_a(\emptyset, \theta_b) = (1 - \lambda)\mu(P; \theta_b) + \lambda\theta_b$, where P is the non-disclosure pool and $\mu(P; \theta_b) = \mathbb{E}[\theta_a | \theta_a \in P, \theta_b]$.

We adopt two standard conventions: (C1) an indifferent sender withholds; (C2) if $P = \emptyset$ (off path), the pool mean is set to the worst state, $\mu = 1$. An *equilibrium* is a pool satisfying $P = \{\theta_a \in \mathcal{A} : \theta_a \leq \hat{\theta}_a(\emptyset, \theta_b)\}$. As the right-hand side is a lower interval, every equilibrium pool is $\{1, \dots, k\}$, and k is the *equilibrium cutoff*, the counterpart of $\theta_a^*(\theta_b)$.

C.2 The non-disclosure pool never contains the public signal

Lemma C1. *In every equilibrium: (i) if $\theta_b = 1$, the equilibrium is unique for all (λ, ρ) , with pool $\{1\}$ and guess $\hat{\theta}_a = 1$; (ii) if $\theta_b \in \{3, 5\}$, the cutoff satisfies $k \leq \theta_b - 1$, so the pool distribution is uniform, $\mu = (k + 1)/2$, independent of ρ .*

Proof. The argument mirrors the proof of Proposition 1 (Appendix B) establishing $\theta_a^*(\theta_b) < \theta_b$. If $k \geq \max\{\theta_b, 2\}$, the pool places probability $q > 0$ on state $1 < k$, so $\mu < k$ and $\hat{\theta}_a \leq (1 - \lambda)\mu + \lambda k < k$, contradicting $k \leq \hat{\theta}_a$. This proves (ii) and rules out $k \geq 2$ at $\theta_b = 1$, thus also proving (i). $\square$

Corollary C1. *The equilibrium set depends on $(\rho, \lambda_\rho, \theta_b)$ only through (λ_ρ, θ_b) : all comparative statics in ρ operate through λ_ρ , as in the continuum. Moreover, if $\lambda = 0$ then, exactly as in Corollary 1, the unique equilibrium at every θ_b has pool $\{1\}$ and guess 1 (full unraveling), since $k \leq \mu = (k + 1)/2$ forces $k = 1$.*

C.3 Characterization

For $\theta_b \in \{3, 5\}$, $k \in \{1, \dots, \theta_b - 1\}$, and $\lambda \in (0, 1)$, let $T(k) \equiv (1 - \lambda)(k + 1)/2 + \lambda\theta_b$ denote the guess induced by pool $\{1, \dots, k\}$, and let $s(\theta_b, \lambda) \equiv [(1 - \lambda) + 2\lambda\theta_b]/(1 + \lambda)$ be the continuum threshold of Equation 3 with $\theta_a^{\min} = 1$, where s solves the fixed-point equation $x = (1 - \lambda)(1 + x)/2 + \lambda\theta_b$.

Proposition C1. Fix $\theta_b \in \{3, 5\}$ and $\lambda \in (0, 1)$. Then k is an equilibrium cutoff iff $k \leq T(k) < k + 1$, equivalently iff $s - 2/(1 + \lambda) < k \leq s$. Consequently: (i) an equilibrium exists for every λ ; (ii) the equilibrium set $E(\theta_b, \lambda)$ consists of at most two consecutive integers; (iii) every equilibrium cutoff lies within $2/(1 + \lambda) < 2$ of s ; (iv) any two coexisting equilibria have cutoffs differing by one and guesses differing by exactly $T(k + 1) - T(k) = (1 - \lambda)/2 < 1/2$.

Proof. By Lemma C1(ii), the discrete equilibrium condition is that type k withholds and type $k + 1$ discloses, $k \leq T(k) < k + 1$. Rearranging as in the proof of Proposition 1, $k \leq T(k) \iff k \leq s$ and $T(k) < k + 1 \iff k > 2\lambda\theta_b/(1 + \lambda) - 1 = s - 2/(1 + \lambda)$. The interval $(s - 2/(1 + \lambda), s]$ has length $2/(1 + \lambda) \in (1, 2)$, hence contains one or two (consecutive) integers. Its largest integer lies in $\{1, \dots, \theta_b - 1\}$ because $1 \leq s < \theta_b$. $\square$

Table A10: Equilibrium cutoff sets $E(\theta_b, \lambda)$ in the discrete game

$\theta_b = 3$		$\theta_b = 5$	
λ range	E	λ range	E
$(0, 1/3)$	$\{1\}$	$(0, 1/7)$	$\{1\}$
$[1/3, 1/2)$	$\{1, 2\}$	$[1/7, 1/4)$	$\{1, 2\}$
$[1/2, 1)$	$\{2\}$	$[1/4, 1/3)$	$\{2\}$
		$[1/3, 3/7)$	$\{2, 3\}$
		$[3/7, 3/5)$	$\{3\}$
		$[3/5, 2/3)$	$\{3, 4\}$
		$[2/3, 1)$	$\{4\}$

C.4 Hypotheses 1 and 2 in the discrete game

In this section, “every equilibrium selection” means an arbitrary element of E at each parameter configuration, with no consistency imposed across configurations. Equation 6 translates the threshold and guess orderings below into the disclosure-rate orderings stated in Hypotheses 1 and 2.

Proposition C2 (Hypothesis 1). Fix $\lambda \in (0, 1)$ and $\rho \in (0, 1)$. For any $\theta_b < \theta'_b$ in $\mathcal{B}$, any $k \in E(\theta_b, \lambda)$, and any $k' \in E(\theta'_b, \lambda)$: $k \leq k'$ and $T(k'; \theta'_b) \geq T(k; \theta_b) + \lambda(\theta'_b - \theta_b) > T(k; \theta_b)$. Hence, across every equilibrium selection, non-disclosure guesses strictly increase in θ_b and withheld sets are nested, so disclosure weakly decreases in θ_b state by state, strictly for $\theta_a \in \{k + 1, \dots, k'\}$ whenever $k' > k$.

Proof. Nestedness: at $\theta_b = 1$, $k = 1 \leq k'$ by Lemma C1(i); for $(\theta_b, \theta'_b) = (3, 5)$, $k' < k$ would force $k = 2$ and $k' = 1$, but $2 \in E(3, \lambda)$ requires $\lambda \geq 1/3$ while $1 \in E(5, \lambda)$ requires $\lambda < 1/4$ (Proposition C1), a contradiction. Guesses: $T(k'; \theta'_b) \geq T(k; \theta'_b) = T(k; \theta_b) + \lambda(\theta'_b - \theta_b)$, using $k' \geq k$ and monotonicity of T in k ; at $\theta_b = 1$ the equilibrium guess is $1 < 1 + \lambda(\theta'_b - 1) \leq T(k'; \theta'_b)$. $\square$

Proposition C3 (Hypothesis 2). (i) At $\theta_b = 1$, the equilibrium pool is $\{1\}$ and the guess equals 1 for all (λ_ρ, ρ) : disclosure and non-disclosure beliefs are exactly invariant to ρ , so $LC = MC = HC$. (ii) At $\theta_b \in \{3, 5\}$, the smallest and largest equilibrium cutoffs, $\underline{k}(\lambda) = \min E(\theta_b, \lambda)$ and $\bar{k}(\lambda) = \max E(\theta_b, \lambda)$, are nondecreasing in λ , and the associated guesses $T(\underline{k}(\lambda); \lambda)$ and $T(\bar{k}(\lambda); \lambda)$ are strictly increasing in λ . Hence, if λ_ρ strictly increases in ρ , then whether one selects the smallest or the largest equilibrium, non-disclosure guesses strictly increase in ρ and disclosure weakly decreases, strictly for intermediate private states whenever λ_ρ crosses a boundary of Table A10; under an arbitrary selection the same orderings hold up to the band of Proposition C1(iv), bounded by $(1 - \lambda)/2 < 1/2$.

Proof. (i) is Lemma C1(i). (ii): s is strictly increasing in λ , the discrete restatement of Corollary 3, and so is the lower endpoint $s - 2/(1 + \lambda) = 2\lambda\theta_b/(1 + \lambda) - 1$. Since $E(\lambda) = \mathbb{Z} \cap (a(\lambda), b(\lambda)]$ with a, b increasing and $b - a > 1$, $\bar{k}(\lambda)$ is the largest integer not exceeding $b(\lambda)$ and $\underline{k}(\lambda)$ the smallest integer exceeding $a(\lambda)$, so both are nondecreasing in λ . For $\lambda_1 < \lambda_2$, $T(\bar{k}(\lambda_2); \lambda_2) \geq T(\bar{k}(\lambda_1); \lambda_2) > T(\bar{k}(\lambda_1); \lambda_1)$, the strict step because $\partial T/\partial \lambda = \theta_b - (k + 1)/2 \geq \theta_b/2 > 0$ for $k \leq \theta_b - 1$; the argument for $\underline{k}$ is identical. $\square$

Remark C1 (Point predictions at the estimated parameters). At $\hat{\lambda}_{0.2} = 0.169$, $\hat{\lambda}_{0.5} = 0.262$, $\hat{\lambda}_{0.8} = 0.320$, Table A10 gives $E(3, \hat{\lambda}) = \{1\}$ in all treatments and $E(5, \hat{\lambda}) = \{1, 2\}$ in LC, $\{2\}$ in MC and HC, with guesses 1 at $\theta_b = 1$; 1.338, 1.524, 1.640 at $\theta_b = 3$; and $\{1.676, 2.092\}$, 2.417, 2.620 at $\theta_b = 5$. Every equilibrium selection therefore satisfies the guess orderings of both hypotheses strictly at the estimated parameters.

C.5 The half-unit guess grid

In the experiment the receiver's action lies in $\mathcal{G} = \{1, 1.5, 2, \dots, 5\}$. We let her play the grid point nearest her target, $r(\hat{\theta}_a)$, with ties rounded up, so r is nondecreasing and $|r(x) - x| \leq 1/4$, and the sender discloses iff $\theta_a > r(\hat{\theta}_a)$.

Observation C1. Rounding can create equilibria in which the rounded guess coincides exactly with an interior withheld type, which is then exactly indifferent: at $\theta_b = 3$ in HC, the pool $\{1, 2, 3\}$ yields the target ≈ 2.911 , which rounds to 3, leaving type 3 indifferent. Such equilibria require the exact coincidence $r(\hat{\theta}_a) = \theta_a$ for a non-minimal type, disappear under the opposite tie-breaking convention, and are not limits of equilibria under finer grids (as the grid step vanishes, Lemma C1 applies). We therefore restrict attention to *strict* equilibria: every type $\theta_a \geq 2$ strictly prefers its equilibrium action, while type 1 may be indifferent and withholds.

Proposition C4 (Grid game). *Restrict attention to strict equilibria and fix $\theta_b \in \{3, 5\}$, $\lambda \in (0, 1)$.*

- (i) *Every equilibrium cutoff satisfies $k \leq \theta_b - 1$, so no withheld state reaches the public signal and Corollary C1 continues to hold: if $k \geq \theta_b$, the target lies strictly below k (proof of Lemma C1), so $r(\hat{\theta}_a) \leq k$, whereas strict withholding by type $k \geq 2$ requires $r(\hat{\theta}_a) \geq k + \frac{1}{2}$.*
- (ii) *$k = 1$ is an equilibrium iff $T(1) < \frac{7}{4}$; $k \in \{2, \dots, \theta_b - 1\}$ is an equilibrium iff $T(k) \in [k + \frac{1}{4}, k + \frac{3}{4})$, with equilibrium guess $k + \frac{1}{2}$.*

- (iii) The equilibrium is unique whenever it exists: the window in (ii) has length $\frac{1}{2}$, smaller than the decrement $(1 + \lambda)/2$ of $T(k) - k$, so at most one $k \geq 2$ qualifies, and $k = 1$ cannot coexist with any $k \geq 2$ since $T(2) \geq \frac{9}{4}$ implies $T(1) = T(2) - \frac{1-\lambda}{2} \geq \frac{7}{4}$.
- (iv) A strict equilibrium exists for all λ outside $[\frac{3}{8}, \frac{1}{2}) \cup [\frac{5}{6}, 1)$ at $\theta_b = 3$ and $[\frac{3}{16}, \frac{3}{14}) \cup [\frac{5}{14}, \frac{5}{12}) \cup [\frac{7}{12}, \frac{7}{10}) \cup [\frac{9}{10}, 1)$ at $\theta_b = 5$; on these exceptional sets only equilibria with an exactly indifferent marginal type (or mixed-strategy equilibria) exist.
- (v) At $\theta_b = 1$ the unique equilibrium guess is exactly 1, invariant to (λ, ρ) .
- (vi) Wherever strict equilibria exist at the compared parameters, all orderings of Propositions C2 and C3 hold weakly, since r is nondecreasing and the unique cutoff is nondecreasing in λ and θ_b (for nestedness: $2 \in E(3)$ requires $\lambda \geq \frac{1}{2}$ while $1 \in E(5)$ requires $\lambda < \frac{3}{16}$); a guess ordering is strict whenever the corresponding target gap exceeds $\frac{1}{2}$, since $y - x > \frac{1}{2}$ implies $r(y) \geq y - \frac{1}{4} > x + \frac{1}{4} \geq r(x)$ —in particular, the cross-signal guess ordering is strict for every selection whenever $\lambda > \frac{1}{4}$, targets differing by at least 2λ across adjacent signals.

Proof. (i) and (v) are immediate. (ii): strict withholding by type $k \geq 2$ requires $r(T(k)) \geq k + \frac{1}{2} \iff T(k) \geq k + \frac{1}{4}$, strict disclosure by type $k + 1$ requires $r(T(k)) \leq k + \frac{1}{2} \iff T(k) < k + \frac{3}{4}$; for $k = 1$, only strict disclosure by type 2 binds, $r(T(1)) < 2 \iff T(1) < \frac{7}{4}$. (iii) as stated. (iv): evaluate (ii) at each k ; e.g., at $\theta_b = 3$, $k = 1 \iff \lambda < \frac{3}{8}$ and $k = 2 \iff \lambda \in [\frac{1}{2}, \frac{5}{6})$, and analogously at $\theta_b = 5$. (vi): the unique k solves $T(k) - k \in [\frac{1}{4}, \frac{3}{4})$ with $T(k) - k$ decreasing in k and increasing in λ and θ_b ; the rest follows from monotonicity of r and the displayed rounding inequality. $\square$

Remark C2 (Grid predictions at the estimated parameters). None of the three $\hat{\lambda}$ falls in an exceptional set, so the grid equilibrium is unique in every cell, with guesses 1 at $\theta_b = 1$ and 1.5 at $\theta_b = 3$ in all treatments, and 1.5 (LC), 2.5 (MC), 2.5 (HC) at $\theta_b = 5$.

D Experimental Instructions

Welcome

You are about to participate in an experiment on decision-making, and you will be paid for your participation in cash, privately at the end of the experiment. What you earn depends partly on your decisions, and partly on chance. Please silence and put away your cellular phones now. The entire session will take place through your computer terminal. Please do not talk or in any way communicate with other participants during the session. We will start with a brief instruction period. During the instruction period, you will be given a description of the main features of the experiment and will be shown how to use the computers. If you have any questions during this period, raise your hand and your question will be answered so everyone can hear.

Instructions

- The experiment you are participating in consists of 50 rounds. At the end of the final round, the computer will select two random rounds, and you will be paid based on your total earnings in those two rounds (in addition to the \$10 show-up fee). The currency used during the experiment is what we call “Experimental Currency Units” (ECU). For your final payment, your earnings in those two rounds will be converted into US dollars at the ratio of 20:1 (20 ECU=\$1). They will then be rounded to the nearest dollar amount. Everybody will be paid in private. You are under no obligation to tell others how much you earned.
- In the first round, you will be matched with one other person, and you are equally likely to be matched with any other person in the room. You will not know whom you are matched with, nor will the person who is matched with you. One of you will be assigned to be the sender (**S Player**) and the other to be the receiver (**R Player**) for that round. You are equally likely to be assigned to either role. In the second round, you will once again be randomly matched with one other person (most likely with a different person than in the first round) and randomly assigned a role, and this will be repeated until 50 rounds are complete.
- In each round, the computer will generate two numbers, A and B. A is drawn from the set $\{1, 2, 3, 4, 5\}$ and B is drawn from the set $\{1, 3, 5\}$. They are generated using the following rule:
 - First, B is randomly drawn from the set $\{1, 3, 5\}$ with equal probability.
 - Then, with 50% chance, A is equal to B (i.e. $A=B$).
with 50% chance, A is randomly drawn from the set $\{1, 2, 3, 4, 5\}$ with equal probability.

Then, for example, if $B=3$, the probability that A is equal to each of $\{1, 2, 3, 4, 5\}$ is given by the following table:

When B=3					
Number A	A=1	A=2	A=3	A=4	A=5
Probability	10%	10%	60%	10%	10%

The computer will send the number B to both the **S Player** and the **R Player**. Then, the computer will send the number A only to the **S Player**. After receiving these numbers, the **S Player** will choose whether or not to report the number A to the **R Player**. If the **S Player** chooses to report the number A, the **R Player** will receive this message from the **S Player**: "The number A I received is" followed by the actual number A. Otherwise, the **R Player** will receive no message, and will only observe the number B.

- After seeing the message or not, the **R Player** will guess the value of the number A. The earnings of both players depend on the value of the number A and the **R Player**'s guess.
- The specific earnings are shown in the table below, which is displayed again before the **S Player** and **R Player** make their choices. In each cell of the table, the payoff for the **S Player** is on the left, and the payoff for the **R Player** is on the right. As you can see from the table, the **S Player** earns more when the **R Player** makes a higher guess on the number A, and the **R Player** earns more when their guess is closer to the number A.
- The next page contains a short quiz, after which the experiment will begin.

PAYOFFS: S, R	Player R's Guess								
	1	1.5	2	2.5	3	3.5	4	4.5	5
A = 1	-29, 110	-6, 102	17, 90	38, 75	57, 57	75, 38	90, 17	102, -6	110, -29
A = 2	-29, 90	-6, 102	17, 110	38, 102	57, 90	75, 75	90, 57	102, 38	110, 17
A = 3	-29, 57	-6, 75	17, 90	38, 102	57, 110	75, 102	90, 90	102, 75	110, 57
A = 4	-29, 17	-6, 38	17, 57	38, 75	57, 90	75, 102	90, 110	102, 102	110, 90
A = 5	-29, -29	-6, -6	17, 17	38, 38	57, 57	75, 75	90, 90	102, 102	110, 110

E Screenshots

Round 1

You are the S Player.

Number A	5
Number B	5

Would you like to report your number A?

- ☐ Report
☐ Do Not Report

Reminder:

When B = 5					
Number A	A = 1	A = 2	A = 3	A = 4	A = 5
Probability	10%	10%	10%	10%	60%

PAYOFFS: S, R	Player R's Guess								
	1	1.5	2	2.5	3	3.5	4	4.5	5
A = 1	-29, 110	-6, 102	17, 90	38, 75	57, 57	75, 38	90, 17	102, -6	110, -29
A = 2	-29, 90	-6, 102	17, 110	38, 102	57, 90	75, 75	90, 57	102, 38	110, 17
A = 3	-29, 57	-6, 75	17, 90	38, 102	57, 110	75, 102	90, 90	102, 75	110, 57
A = 4	-29, 17	-6, 38	17, 57	38, 75	57, 90	75, 102	90, 110	102, 102	110, 90
A = 5	-29, -29	-6, -6	17, 17	38, 38	57, 57	75, 75	90, 90	102, 102	110, 110

Next

Figure E.1: Screenshot of Sender Page

Round 1

You are the R Player.

Number A	?
Number B	5
Message from the S Player	"No message."

What is your best guess of the number A?

- ☐ 1
☐ 1.5
☐ 2
☐ 2.5
☐ 3
☐ 3.5
☐ 4
☐ 4.5
☐ 5

Reminder:

When B = 5					
Number A	A = 1	A = 2	A = 3	A = 4	A = 5
Probability	10%	10%	10%	10%	60%

PAYOFFS: S, R	Player R's Guess								
	1	1.5	2	2.5	3	3.5	4	4.5	5
A = 1	-29, 110	-6, 102	17, 90	38, 75	57, 57	75, 38	90, 17	102, -6	110, -29
A = 2	-29, 90	-6, 102	17, 110	38, 102	57, 90	75, 75	90, 57	102, 38	110, 17
A = 3	-29, 57	-6, 75	17, 90	38, 102	57, 110	75, 102	90, 90	102, 75	110, 57
A = 4	-29, 17	-6, 38	17, 57	38, 75	57, 90	75, 102	90, 110	102, 102	110, 90
A = 5	-29, -29	-6, -6	17, 17	38, 38	57, 57	75, 75	90, 90	102, 102	110, 110

Next

Figure E.2: Screenshot of Receiver Page

Additional Questions

Please answer the following questions carefully.

Suppose the S Player's number B is 1.

What percentage of S Players do you think reported the number A? [Please select a number from 0 to 100]

When the number A was 1: %

When the number A was 2: %

When the number A was 3: %

When the number A was 4: %

When the number A was 5: %

What do you think the average R Player's guess was when the number was not reported? [Please select a number between 1 and 5]

Suppose the S Player's number B is 3.

What percentage of S Players do you think reported the number A? [Please select a number from 0 to 100]

When the number A was 1: %

When the number A was 2: %

When the number A was 3: %

When the number A was 4: %

When the number A was 5: %

What do you think the average R Player's guess was when the number was not reported? [Please select a number between 1 and 5]

Suppose the S Player's number B is 5.

What percentage of S Players do you think reported the number A? [Please select a number from 0 to 100]

When the number A was 1: %

When the number A was 2: %

When the number A was 3: %

When the number A was 4: %

When the number A was 5: %

What do you think the average R Player's guess was when the number was not reported? [Please select a number between 1 and 5]

Next

Figure E.3: Screenshot of Belief Questions